

A-Z OF SCIENCE

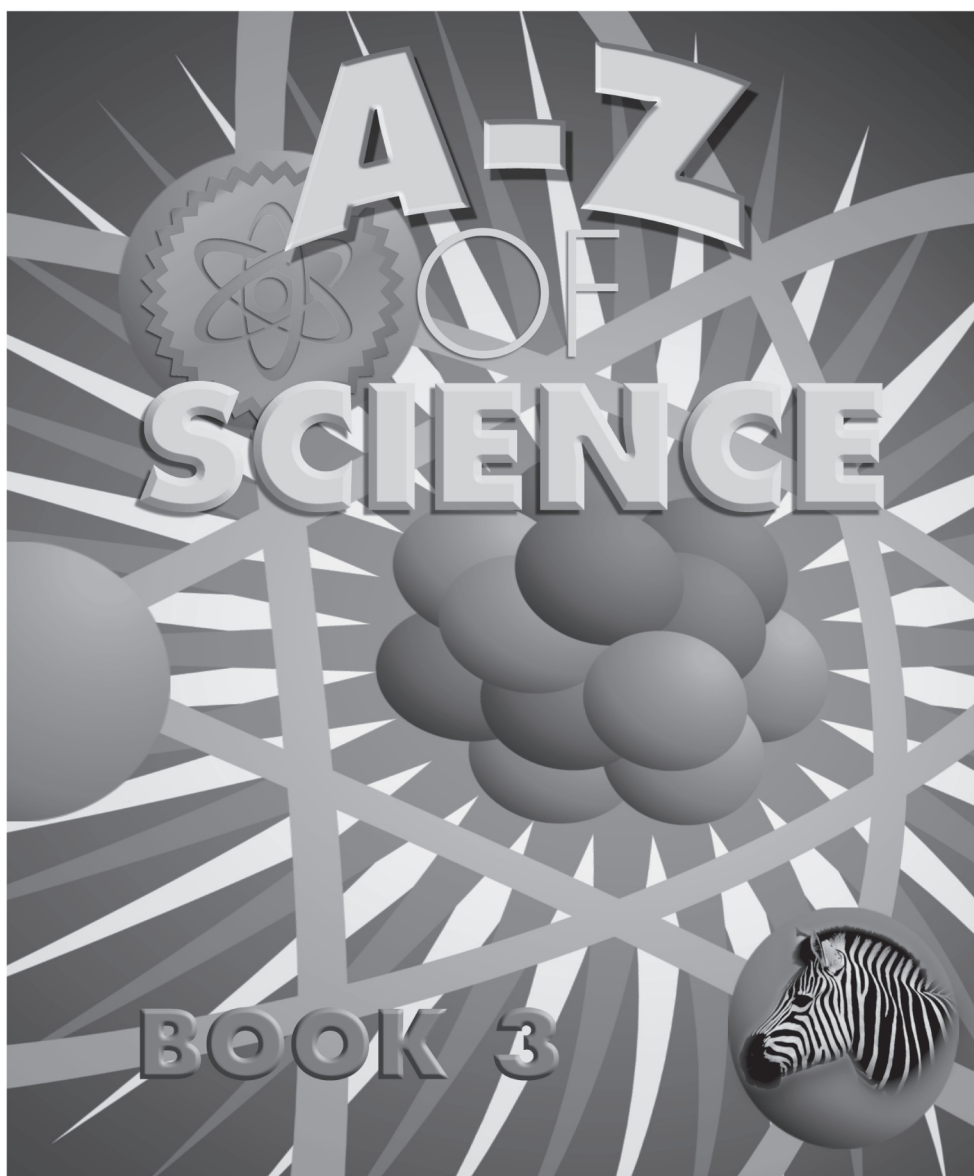
BOOK 3

Zenda Spry-Pohl and
Raimund Pohl



Quality ready-to-use
resources





Zenda Spry-Pohl and
Raimund Pohl

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Introduction

This series of photocopiable books extends across all branches of this subject area, including biology, chemistry, physics, astronomy, earth and environmental science. The activities include basic revision of fundamental concepts as well as ideas and tasks that will challenge students to think beyond recalling facts. Each topic is designed to engage student interest in science, as well as consolidate and extend their scientific knowledge and understanding. Book 3 is suitable for years 7 and 8, while Book 4 is more advanced and will suit years 9 and 10. Teachers will find these worksheets useful in a variety of ways, including homework, assessment, revision and extension. They are designed to suit both independent and group work, and are also convenient ways of targeting the learning of individual students. For some worksheets, students may find textbooks or research materials useful.

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Adaptations

An **adaptation** is a characteristic that suits a plant or animal to its particular environment.

Kangaroos around Darwin in the north of Australia seek shade in rocky outcrops and they avoid moving around in the heat of the day. They have large, thin ears so they can lose heat more easily because the warm blood in the ears dissipates its heat to the surrounding air.



Agile wallaby

1 List three other animals that have large ears to help them lose heat easily.

Trees in the Daintree Rainforest in northern Queensland can withstand large amounts of rain. This is because they have efficient ways of removing water – their leaves have drip tips that let the rain run off. They have also adapted to the low light levels by developing large, dark green leaves that have lots of the pigment chlorophyll. This allows efficient photosynthesis.

2 In the box at the right, draw and label a common house plant that does not need much light.

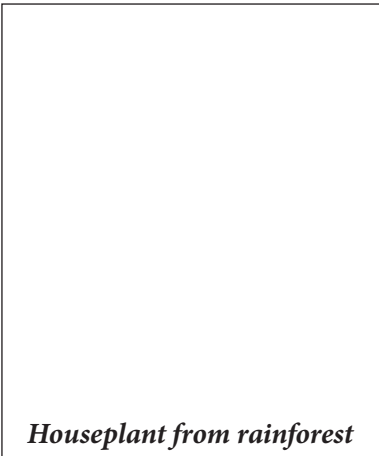
3 Even the grasshoppers in your backyard have adaptations. Describe two adaptations of grasshoppers and their uses.

a Adaptation: _____

Use: _____

b Adaptation: _____

Use: _____



Houseplant from rainforest

4 Write the correct adaptation and function beside each organism in the table below.

ADAPTATIONS: thick seed cones, pointed long beak, sticky hairs, flexible tail, strong front claws

FUNCTIONS: to grab lizards, to dig, to catch insects, to climb trees, to resist fire

ORGANISM	ADAPTATION	FUNCTION
possum		
banksia tree		
wombat		
sundew		
kookaburra		

Structural and behavioural adaptations

An adaptation can be part of the structure of an organism's body, or it can be a behaviour the organism shows.

5 Use the words to write clear sentences describing each organism's adaptation.

a Crab: hard shell, stops it being eaten, structure, hides in holes, keeps out of sight of predators, behaviour

b Emu: long thin legs, fast, pecks strongly, resists predators, structure, behaviour

c Platypus: water-resistant fur, warm, structure, swims near banks of creeks, worms and larvae, behaviour

Adaptations of organisms

6 Use the list of organisms in the box below to complete the first column of the table below. Then complete the other two columns from your general knowledge.

Tip!

Give more information in your answer by using an ADJECTIVE with the NOUN, e.g. pointed beak.

+ bilby	+ mangrove	+ coral	+ coral fish	+ gecko
+ eucalypt tree	+ koala	+ Tasmanian devil	+ grass tree	+ pitcher plant
+ euro or red wallaroo	+ wattle	+ green tree frog	+ rosella parrot	+ coral
+ crocodile				

Predators a List four organisms and the adaptation each one has to enable it to catch or eat food.		
ORGANISM	METHOD OF GAINING FOOD	TYPE OF FOOD
pelican		
Prey b List four organisms and the adaptation each one has for protection. Think about camouflage, speed, strong shells and poisons.		
ORGANISM	TYPE OF PROTECTION	PROTECTION FROM
hopping mouse		
Climate c List four organisms and the adaptation each one has for living in a harsh Australian climate.		
ORGANISM	TYPE OF PROTECTION	DESCRIPTION OF CLIMATE
bottlebrush tree	small, tough leaves	low rainfall, high temperature

Botany

Botany is the study of plants. Plants can be placed into five phyla, or groups, which are shown below.

 Algae: • single cells • no stems • no leaves	 Mosses: • small leaves • no stems • spores	 Ferns: • fronds • stems • spores	 Conifers: • pine needles • stems, roots • seeds in cones	 Flowering plants: • flowers • stems, roots • seeds in fruit
---	--	--	---	---

1 Write the common name of an example for each group in the table. Then name and describe a place where you have seen each plant.

GROUP	COMMON NAME OF EXAMPLE	LOCATION
algae		
mosses		
ferns		
conifers		
flowering plants		

Students and their plant experiences

- 2** You be the judge! In the table on the next page, write the name of the plant group that matches each description. Then list two reasons for each group you chose. The first one has been done for you.
- a** Kim is walking in the jungle and finds amazing small plants with tiny green leaves growing up a tree. They have little brown spore capsules on the ends of the plant.
- b** There is a tree in the school grounds that looks like a Christmas tree. It has needles and sometimes you can see seeds in the opening cones.
- c** Mai can see green slimy stuff growing in the water around the edges of the pond at the park. The tadpoles seem to be eating it.
- d** While walking near a creek in a warm gully, Julie sees plants taller than she is. They have feathery leaves with little brown dots underneath, but no flowers.
- e** Hazim has an orange tree in his backyard that has many white flowers on it. He planted it two years ago from some seeds he found in an orange he was eating.
- f** The grass brushes Maria’s legs as she walks along. When she bends down to grab it, she notices seeds like grains of wheat on the stalk.
- g** Jim’s dad is always complaining about the green stuff in the lawn that looks like miniature plants that have no stalks or seeds. He can’t get the grass to grow there.
- h** At the beach, Kamlesh and Bonnie see great masses of green slimy stuff washed up on the shore. They can’t see any flowers or stalks. It smells of rotting vegetation.

PLANT GROUP	REASON 1	REASON 2
a moss	small leaves	spore capsules
b		
c		
d		
e		
f		
g		
h		

How plants make food

Plants make their own food by a process known as **photosynthesis**. *Photo* means ‘light’ and *synthesis* means ‘make by putting things together’. Plants put together carbon dioxide and water to make oxygen and glucose. Glucose is plant ‘food’. This process can only happen in light, generally sunlight.

- 3 Complete this word formula.

+

→

glucose +
- 4 Where does the carbon dioxide come from?
- 5 Where does the oxygen go?
- 6 List five foods you eat that come from plants.

Fast growing plants

Some plants photosynthesise rapidly, so they grow fast.

There are several ways to measure ‘fast’, such as growth in height, horizontal spread, reproduction rate, dry weight gain, and life cycle. Bamboo is often considered the fastest growing plant, height wise. It has been reported that it may grow up to one metre in height in just one day. During peak growth, sweet corn can gain about 50 grams of dry matter per square metre per day.

Rocket is a fast plant, which can complete its seed-to-seed life cycle in about five weeks under a fluorescent light bank in a classroom. Try growing some in class or at home for a measurement project.

Many weeds are very rapid growers. Two aquatic weeds that spread extremely rapidly are water hyacinth and giant salvinia.

- 7 Describe the fastest growing plant in your neighbourhood. List the benefits and problems associated with growing this plant. Use this template to help you organise your observations.

Name of plant:	Plant group:
Description of the plant:	It is found in:
It can be a problem because:	Benefits of growing it are:
People remove it by:	Conclusion:

Cells

1 Choose the correct words from the box below to fill in the blanks.

+ organelles	+ covering	+ space	+ holds	+ through	+ drying
+ common	+ nucleus	+ structures	+ chloroplasts	+ jelly	+ moving

All living things are made of cells. The cells of plants and animals have some _____ features and some different ones.

The very small structures in a cell are called organelles. All cells have these _____: a nucleus, cytoplasm, and a cell membrane. However, only plant cells have a cell wall, _____ and vacuoles.

The function of the _____ is to control the cell. Although it cannot think, the nucleus controls all the cell's jobs like growing, _____, reproducing and eating. The cell membrane or skin of the cell _____ the cell together. Food, water, oxygen and waste all pass _____ this membrane by a process called diffusion. Cytoplasm is the inside of the cell. It is the fluid of the cell body and is like thin _____.

The cell wall of plants is an extra _____ made of tough cellulose that helps the plant to keep its shape and stops it _____ out too quickly.

Most plant cells contain small green _____, which are food-making organelles called chloroplasts. Animals do not have chloroplasts.

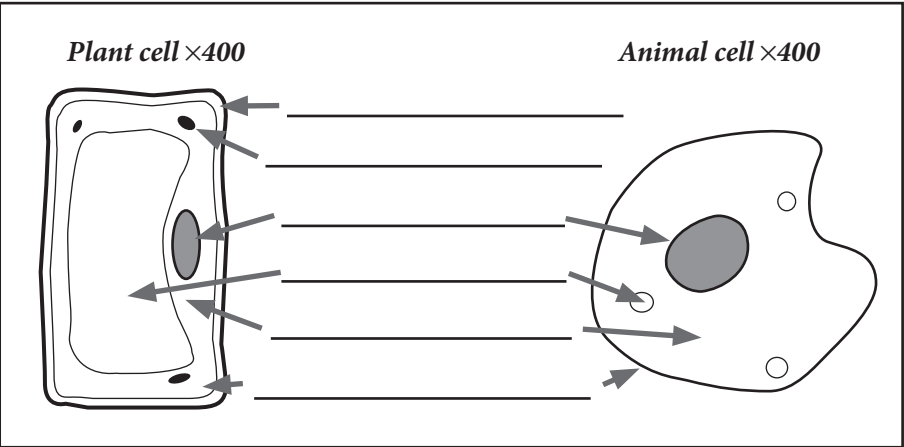
Sometimes cells have a small storage _____ filled with water. This is called a vacuole.

Parts of the cell

2 Complete the table which lists the parts of the cell that you can see through an ordinary light microscope.

ORGANELLE	CAN BE COMPARED TO	WHAT IT DOES
cell membrane	skin	
cell wall		
cytoplasm		allows all the fluids and nutrients to have a base
nucleus		
chloroplast		
vacuole		

3 Label the cells below using these names: cell membrane, cell wall, nucleus, cytoplasm, chloroplast, vacuole.



- 4 Choose one of the tasks below and complete it in your workbook.
- a Write a motivating paragraph to encourage someone to become a cell biologist.
 - b List 10 ways that cells affect your life.

Data gathering and data use

- 1 Gather some data now by asking 10 people to name their favourite confectionary or sweet. Give them a choice by writing four names of popular sweets in the 'Name' row of the table below. Ask each person to mark on your page the one they like best. Use tally marks like this:

|||| to show 4 ||||| to show 5

	CHOICE 1	CHOICE 2	CHOICE 3	CHOICE 4
Name				
Tally				

- 2 Make a column graph of your results. Then write a conclusion in the box.



Conclusion:

Instruments

Data gathering is usually much more detailed than this and uses data loggers or instruments. Real time data is gathered as the event takes place. Data is the difference between an accurate result and guesswork.

- 3 Write the correct instrument beside each type of data needed.

Instrument list			
+ thermosensor	+ pH meter	+ expendable primary	+ flow meter
+ thermometer	+ nitrogen probe	element with data	+ heart rate sensor
+ thermoanemometer	+ chloride probe	transmitting assembly	+ pulse oximeter

DATA NEEDED	INSTRUMENT USED
continuous heat readings	
heart rate	
oxygen level in blood	
wind power from heat readings	
temperature of a molten metal bath	
temperature of water	
acidity of water	
stream flows	
chlorine ions	
nitrate ions	

4 Answer the following questions using the information in the article at the right.

a Identify the new technology and the group that has recently purchased this equipment.

b Describe the type of data gathered by the technology.

c Where are the Doppler radars located?

d Recommend another non-weather use for this technology.

e Explain how the Doppler effect can be used to predict approaching hailstorms.

f List four important items to gather before a serious storm.

If only we knew it was coming ...

Eastern Herald, 30 December 2007

Twenty years ago this was a common cry from people caught in super storms. They didn't even have time to collect their emergency radio, torch, first aid kit and water.

Now with the new weather radars at the Bureau of Meteorology's bases in Adelaide, Melbourne, Sydney and Brisbane, we are able to predict strong weather events.

The new Doppler radars

work on the principle that waves coming towards you bunch up and appear to get faster, and those going away seem to spread out and go slower. They show how fast storms are approaching.

A forecaster from the bureau said the high-resolution Doppler radars would even allow them to track smoke from bushfires and see inside thunderstorms.

Ocean data

Many people live and work by the ocean, and tsunamis and cyclones coming from the ocean have caused many deaths. If we have data on approaching storms, much disaster could be prevented.

Data on tides, wave speed, height and direction can be used to predict ocean weather. Australia has the Bureau of Meteorology (BOM) for collecting data, and the USA has the National Geophysical Data Center.

5 Describe the most vivid ocean disaster that you have heard about. How could the disaster have been prevented?

Use of data

6 Explain how you would manage a possible flood or other serious natural event in your local area. Include the data you would want to know and when you would need to receive this information.

• Measures for saving human lives

• Caring for animals

• Preserving property

• Data needed and time needed

Ecosystems

Earth is a special place where organisms live in unique environments. The organisms require both living (biotic) and non-living (abiotic) components to survive. When two or more organisms live and interact with each other, they form a community. Each community lives in a specific area called a habitat. A habitat contains both biotic and abiotic features. Each specific area with its organisms is called an ecosystem.



- 1** Classify these words into BIOTIC and ABIOTIC by writing them in the correct row below: sunlight, grass, temperature, bark, leaves, chemicals, water, wombat, minerals, caterpillar, oxygen.

BIOTIC	
ABIOTIC	

- 2** In your own words, define ECOSYSTEM.

An ecosystem is _____

- 3** List three different ecosystems in your local area and briefly describe each one.

- a** _____
- b** _____
- c** _____

Each plant and animal has a specific function or position within its community. This function or position is called a niche. Green plants produce their own food and so are called producers. Animals that eat producers are called consumers. There are several levels of consumer. First order consumers eat plants and are herbivores. Second order consumers eat animals and are carnivores. Top order consumers are also carnivores. When a plant or animal dies, it decays. The organisms that break down the plants or animals are bacteria, fungi, worms and other invertebrates. These organisms are called decomposers.

- 4** Write each organism in the correct column: wedge-tail eagle, bottlebrush, algae, fungus, cockroach, witchetty grub, frog, bacteria, grass.

PRODUCER	CONSUMER	DECOMPOSER

Every organism is adapted to its environment. For example, cacti live in arid areas and hold water. An adaptation helps the organism to survive in its habitat by obtaining food and water and protecting itself. Eucalyptus and wattle trees store oil in their leaves to conserve water. When a bushfire occurs, the eucalypts have buds under their bark so that after the fire they can regenerate. Wattles and banksias shed their seed as a fire goes through the bush. Australian plants also use minimal amounts of nutrients and this helps the plants to survive drought and flood. All of these adaptations are survival adaptations.

- 5** You are a National Parks and Wildlife ranger, in charge of 200 hectares of bushland that has recently been burnt. You need to implement a plan to ensure that the area will regenerate properly. Write your plan from just after the fire and for the next three years, in the space on the next page.

Regeneration plan

Immediate action

Consider injured animals, birds and people

Data collection (rainfall, vegetation, animals, birds)

Three year action plan

Year 1

Year 2

Year 3

As organisms are dependent on other organisms, an arrow is used to represent the energy flow between them. For example: *producer* → *consumer* → *consumer 2*. This flow of energy is called a food chain. When a number of food chains are joined together, they form a food web. A food web shows exactly what eats what. A food chain always starts with a green plant or producer.

- 6
- a

Show a food chain by writing each organism in the correct box: wildcat, plants, caterpillar, eagle, small bird.
- b

Label each organism using the terms ‘producer’ and ‘consumer 1, 2, 3 and 4’.

→

→

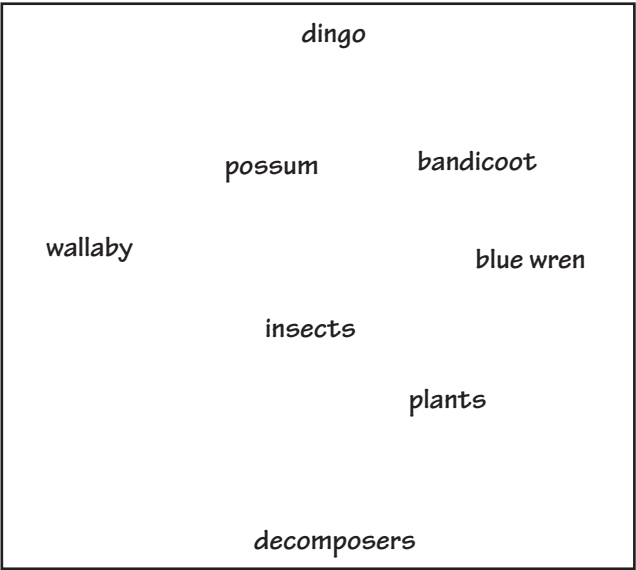
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- 7
- Complete the food web diagram at the right by inserting arrows between the organisms. Remember, arrows always point in the direction of the energy flow.

Plants trap the Sun’s energy with chlorophyll and use it to make their food and oxygen. To make the food, plants absorb carbon dioxide gas from the air and water from the ground. The food that they make is glucose and this is changed into starch. The oxygen produced by the plant is either released into the atmosphere or used to release energy from glucose.

When animals respire, their body cells combine the sugar or food with oxygen to provide the energy they need. At the same time, they release the products carbon dioxide and water. Some of the features of respiration seem like the reverse of photosynthesis.



- 8
- Complete the word equations below.
- a

Photosynthesis: carbon dioxide + water

→

sunlight/energy

→

+
- b

Respiration: glucose +

→

+

water

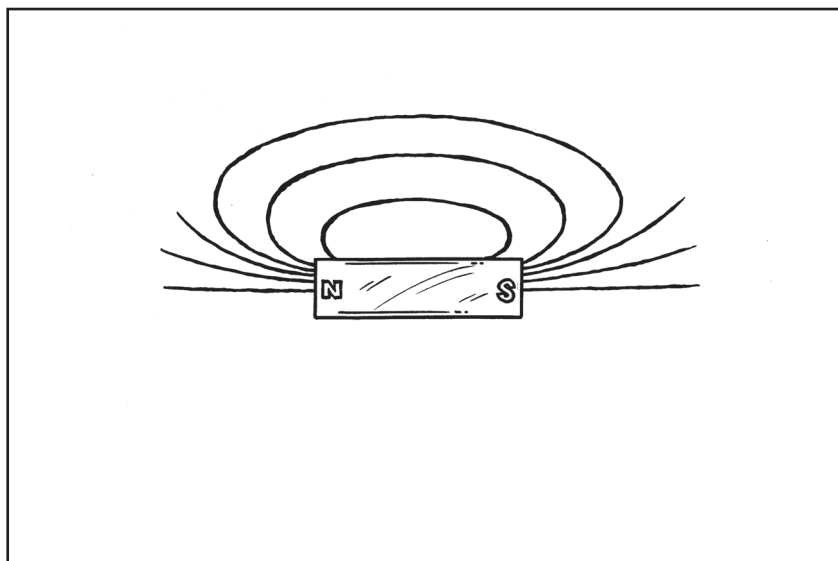
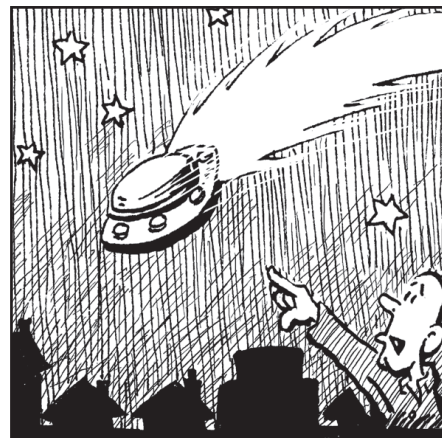
+

Forces

Forces keep you on Earth. They allow you to float on water, and stop you from slipping as you walk. Forces impact on you every day.

Can you imagine a force field around objects? In science fiction programs and video games, there is often a force field around the spaceship. There are also fields around magnets and around the Earth.

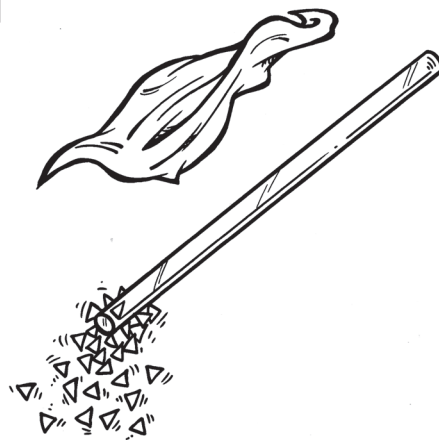
- 1 Invent and draw a force field for the spaceship at the right.
- 2 Complete the magnetic field around this bar magnet.



Earth's gravity extends an exceedingly long way. For practical purposes, space is defined as starting 100 km above Earth. Above this, objects float or keep moving in the direction that they are pushed.

Electrostatic force occurs when you rub an insulated rod and pick up non-metal objects, as you can see in the diagram at the right.

Exploring electrostatic force was the beginning of the invention of electricity. The difficulties in making moving electricity were making these electrons bump the next ones along the wire, producing a current and then controlling that current.



Gravitational force

- 3 What will happen to objects in a spaceship after it leaves Earth's gravity? Why?

- 4 Describe how you think an astronaut would eat these foods when there is no gravity.

a juice _____

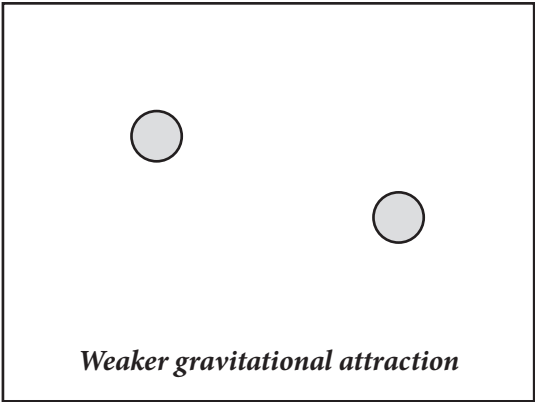
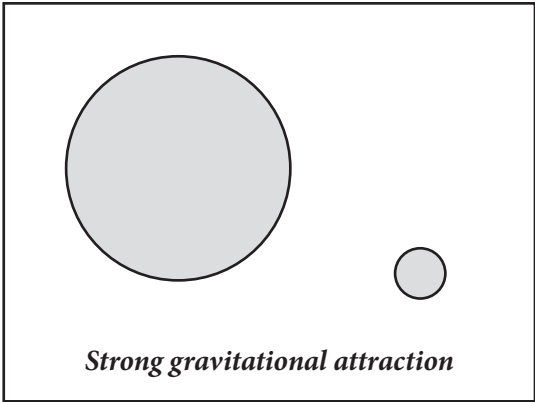
b meat _____

c vegetables _____

d pudding _____

Gravity facts

- + The gravity from the Sun keeps Earth in orbit. The Moon is in orbit around Earth.
- + Every object exerts a force on other objects.
- + Gravity varies with the closeness to the object and the size of the object.



Contact and non-contact forces

Ships float on water when they are in direct contact with the water – they are buoyed up by it. Parachutes glide to Earth slowly when they catch the wind, as they are in direct contact with the wind.

However magnets, electrostatics and gravity do not have to touch an object in order to exert a force on it.

5 List an example of each force acting. Then identify it as either a contact force or a non-contact force.

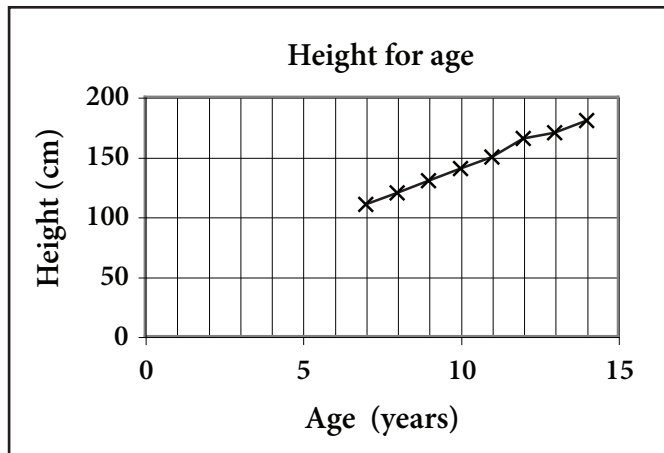
FORCE	EXAMPLE OF ACTION	CONTACT FORCE OR NON-CONTACT FORCE?
magnetism		
electrostatics		
gravity		
buoyancy		
air resistance		

6 List the forces that are in effect in each situation. Then describe how you could resolve each situation and the equipment you would need.

- a Skydiving out of a plane
- b Picking up paperclips that have dropped behind a desk
- c Collecting mini-polystyrene balls from the floor
- d Taking cargo to New Zealand on a ship

Graphing

Graphs are an informative, visual way of viewing data. They can also draw your attention to the facts and figures, if they are presented appropriately. Especially when you have a large amount of information, presenting it as a graph can be useful because you can see the patterns in the information at a glance, and the data can be interpreted more easily.



AGE (yrs)	HEIGHT (cm)
7	110
8	120
9	130
10	140
11	150
12	165
13	170
14	180

- 1 a** Describe the information that you can infer from the graph above.

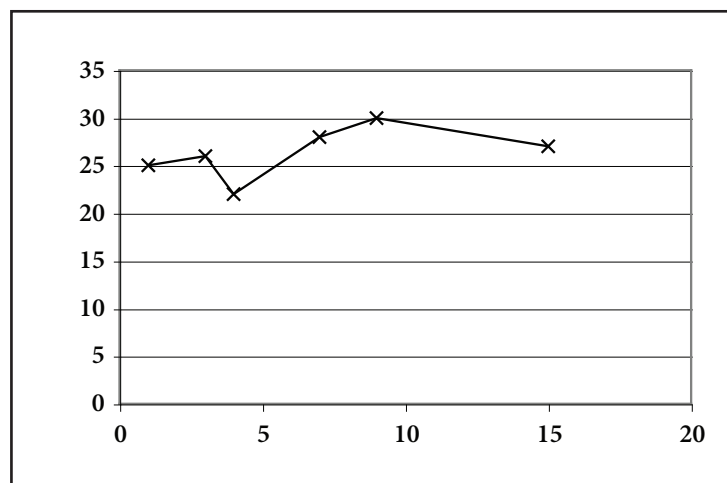
- b** Is it easier to interpret the information from the graph or from the table? Explain why.

Graphing points

On a graph you need to write a heading or title that summarises the factors being graphed. You also need to write the units that are being used.

The numbers on the lines of the graph must always be in regular intervals; e.g. 2, 4, 6 and 8 years, with the same amount of space between each interval. Another example of regular intervals is 1000, 2000, 3000, 4000 steps.

DAY	TEMPERATURE (°C)
1	25
3	26
4	22
7	28
9	30
15	27



- 2** Describe two things that are incorrect in the graph above. Then make corrections to the graph.

- a** _____
- b** _____

Independent axis and dependent axis

Independent axis

The horizontal axis goes across the page and is called the *x*-axis. To remember this, think of the *x*-axis as a cross (X) because it goes across the page. The *x*-axis is where you write independent information like time. Time keeps progressing no matter what happens. It is independent of other factors.

Dependent axis

The vertical axis or *y*-axis is the dependent axis because it depends on the other factor. So your height increases as time goes along until you reach maturity; height depends on time.

- 3 Circle the words that tell you which information should go on the *x*-axis. Then underline the words that tell you which information should go on the *y*-axis.

a The girls' heights change, as they grow older.

b The boys' running speeds vary as the years pass.

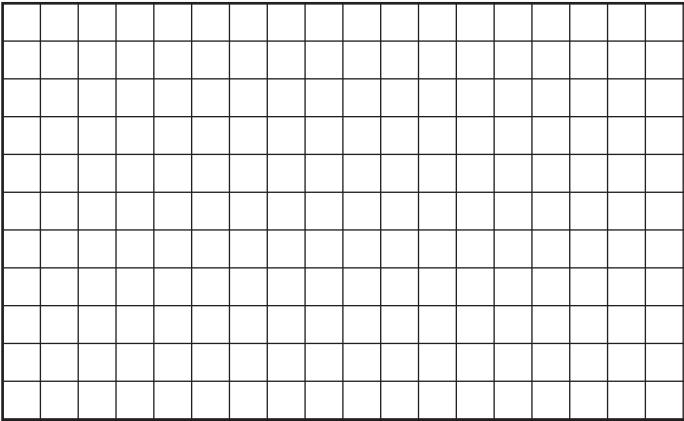
c The amount of petrol the cars use relates to the engine capacity.

d The lung capacity of teenagers can usually be correlated with their height.

- 4 a Draw a graph below using the information in the table.

WATER TEMPERATURE (°C)	BREATHS PER MINUTE
10	15
15	20
20	24
25	28
30	30

EFFECT OF TEMPERATURE ON THE BREATHING RATE OF FISH

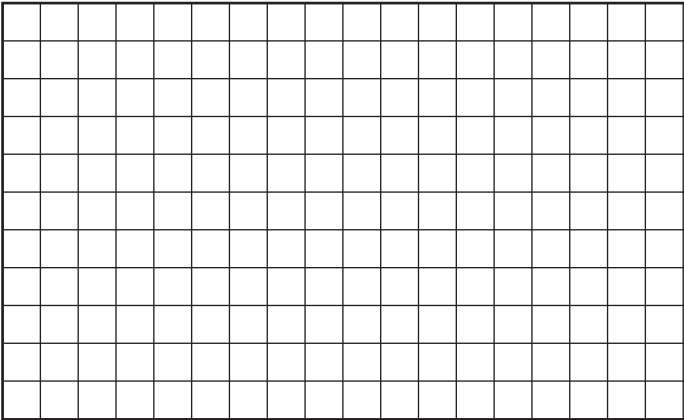


- b What conclusion can you draw from the information in this graph?
- c Infer what the breathing rate will be if the temperature goes higher.

- 5 a Draw a graph below using the information in the table.

TIME RUNNING (MINS)	HEARTBEAT PER MIN
0	70
5	78
10	95
15	130
20	130
25	120
30	100

EFFECT OF RUNNING ON HUMAN HEARTBEAT



- b What conclusion can you draw from the information in this graph?
- c State two good reasons for showing your results table as a graph.

Hydrosphere

Our planet's surface is covered with 70% water. All the surface water is called the hydrosphere, and this is what makes Earth appear blue from outer space. Without water, there is no life.

- 1 Use these words to complete the sentences below: gas, potable, evaporation, surface, cycle, droplets, faster, temperature, energy, precipitation, snow.

The hydrosphere is made up of 97% salt water, 2% ice and 1% fresh water that is suitable for drinking and called _____ water. The water on the Earth is always in motion, as warm water has more _____ and the particles move _____. The water moves around in a _____ because of the Sun's energy.

The Sun warms the _____ layers of the oceans, turning the liquid into a _____, a process called _____. The water vapour rises into the atmosphere until it becomes cool enough to form _____ that make clouds. This process is called condensation. The droplets attach themselves to fine particles and will remain as a cloud until the _____ and pressure changes cause the particles to fall. This is termed _____ and may be hail, _____, sleet or rain.

- 2 Draw a diagram of the water cycle, starting with the waterfall below. Use these labels in your diagram and draw arrows between them to show the direction of the cycle: transpiration, evaporation, condensation, runoff, groundwater, river, ocean, precipitation.



The tides

- 3 Use these words to complete the sentences: hydrosphere, gravitational, Moon, water, same, side, right, six.

The hydrosphere is affected by the _____ pull of the Moon and Sun, and by the Earth's rotation. When the Sun, _____ and Earth are in a line, then the gravitational force causes very high tides, called spring tides. They are called this because the _____ appears to 'spring up' in harbours. There is a full Moon when the Moon is opposite the Sun, and a new Moon when the Moon is on the _____ as the Sun. When the Sun, Moon and Earth are at _____ angles, then very low tides called neap tides occur. Tides change about every _____ hours. The Moon has the major gravitational pull on the _____.

- 4 Look carefully at the diagram at the right and relabel the letters A, B, C and D by writing the terms 'spring tide' and 'neap tide'.

How can the tides help me?

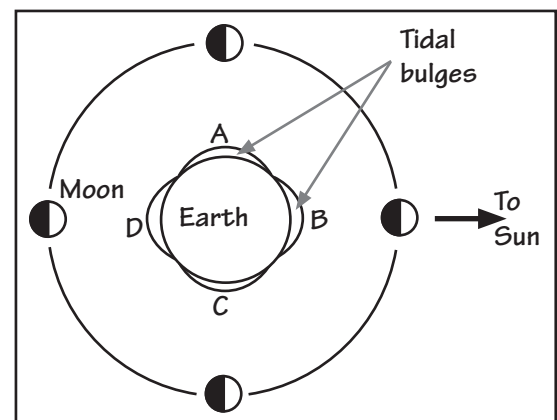
Fishing, surfing and sailing are sports that need tidal information.

- 5 Imagine the water rushing out of the river mouth as a king tide (a large spring tide) goes out. Predict what will happen to each of the following.

a a little fish _____

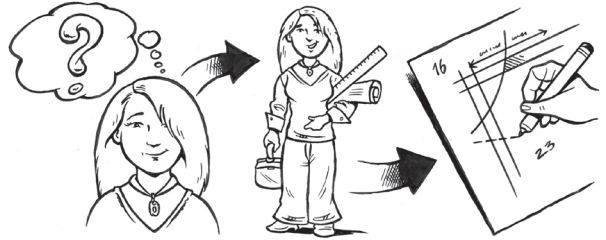
b the surfing waves _____

c a sailing boat leaving harbour _____



Investigations

If you find out something for yourself by doing an experiment, it is called a first-hand investigation. If you read about someone else's work, it is called a second-hand investigation. First-hand information is most valued because it shows you are enterprising, and that you have the real facts if you do the experiment carefully.



- 1** Write 'first-hand investigation' or 'second-hand investigation' beside each statement.
 - a** You weigh two lots of apples to see which is heavier. _____
 - b** You find some figures for the weights of cars from the Internet. _____
 - c** You put salt into four test tubes to watch it dissolve. _____
 - d** You ask your brother to measure three students' heights for you. _____

Setting out an experiment

The following layout is a method for reporting on an experiment.

- 2** Conduct an experiment to see if salt or sugar dissolves better in water. Use the boxes below to help you plan your experiment, and fill them in to record what you find out.

TITLE OF EXPERIMENT:	
Your name:	Date:
HYPOTHESIS I think that _____ _____	
EQUIPMENT: List all the equipment you use. _____ _____	
SAFETY: List all the safety considerations of the experiment. _____ _____	
METHOD _____ _____ _____ _____	

VARIABLES: State the things that could have varied and how you kept them the same. For example, used the same amount of water in each cup.

RESULTS: A table showing the result of each trial.

	TRIAL 1	TRIAL 2	TRIAL 3
Salt			
Sugar			

FEEDBACK: Show your work to an expert in the field or your science teacher or read it to your class. List the feedback you receive.

IMPROVED EXPERIMENT: What you did the second time you conducted the experiment.

RESULTS: Show your results as mathematical data and as a graph. When you draw graphs, say what each one means.

DISCUSSION: Write about what your results could mean. Refer to your results and any information from research.

CONCLUSION: A short concise answer to your hypothesis; a summary of your results; what your results infer.

BIBLIOGRAPHY: Write out clearly the books, websites, journals or other research materials you used. For example:
Kneebone, P.A., (2006). *Skeleton connections*. Body Press, Sydney.
Purkinje, C.E.L., (2007). www.brainsurgery.org.au

Joints

Your body has 206 bones which are linked by joints. This is your **skeletal system**. The bones work like levers to give more speed and power to your movements. The joints allow articulation (movement) of the bones. Ligaments hold the bones together.

Joints like the elbow and ankle allow movement backwards and forwards, like a hinge. The shoulder and hip areas are ball and socket joints that allow rotation. The wrist allows movement in an ellipse in two directions, while the foot bones glide past each other on the same plane. The atlas and axis at the top of your neck allow you to pivot your head in most directions.

1 Complete the following table by extracting the information from the description above.

NAME OF JOINT	PLACE IN BODY	TYPE OF MOVEMENT
ball and socket		
hinge		
pivot		
gliding		
ellipsoid		



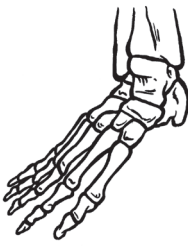
Hip



Elbow



Atlas, axis



Foot



Wrist

2 On each diagram above, draw arrows to indicate the possible movement.

The growth plate and fractures

3 Use these words to complete the following passage: adolescents, long, plate, bone, complete, weakest, solid, boys, damage, fractures, mature, connect.

The growth plate is the area of growing tissue near each end of the _____ bones in children and _____. The growth _____ determines the future length and shape of the mature _____. When growth is _____, sometime during adolescence, the growth plates close and are replaced by bone.

The growth plate is the _____ area of the growing skeleton, weaker than the nearby ligaments and tendons that _____ bones to other bones and muscles. In a growing child, a serious injury to a joint is more likely to _____ a growth plate than the ligaments that stabilise the joint.

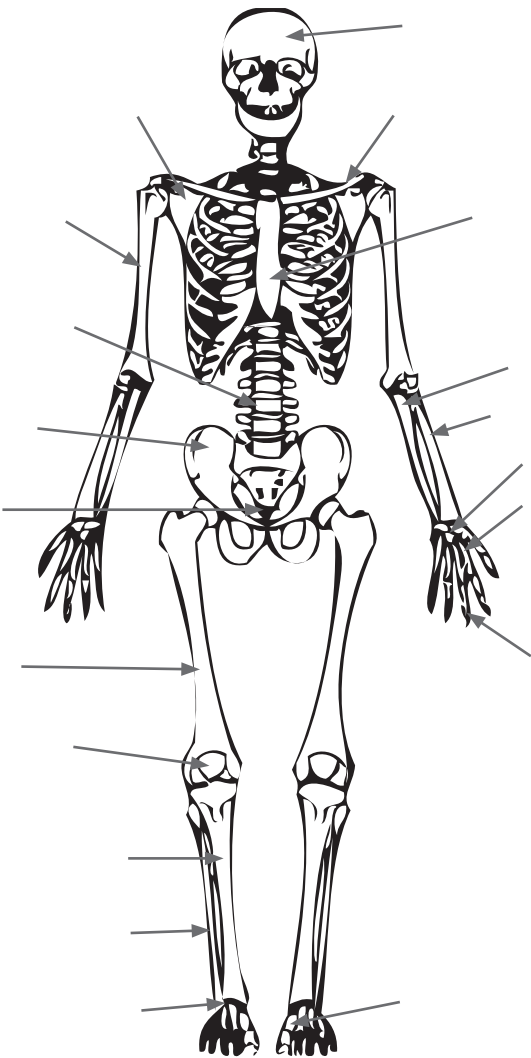
Injuries to growth plates are fractures. They comprise 15% of all childhood _____. They occur twice as often in _____ as in girls, with the greatest incidence among boys aged 14 to 16 years and girls aged 11 to 13 years. Older girls experience these fractures less often because their bodies _____ at an earlier age than boys. This means that their bones finish growing sooner, and their growth plates are replaced by stronger, _____ bone.

- 4** From your experience, describe a fracture that has happened to you or a friend.
- How did the fracture occur? _____
 - What bone was fractured? _____
 - Exact place of fracture on the bone _____
 - Age at time of injury _____
 - Length of time to heal _____

The skeleton

- 5** Use the scientific names in the table below to label each bone marked on the skeleton.

SCIENTIFIC NAME	COMMON NAME
cranium	skull
clavicle	collarbone
scapula	breastbone
radius	lower arm bone
ulna	lower arm bone (above the thumb)
carpels	wrist bones
metacarpals	hand bones
phalanges	fingers
metatarsals	foot bones
scapula	shoulder blade (located behind ribs)
humerus	upper arm bone
vertebrae	backbone
pelvis	hips
coccyx	tail bone
femur	upper leg bone
patella	knee cap
tibia	lower leg (thick bone)
fibula	lower leg (thin bone)
tarsals	ankle bones



- 6** On the skeleton, neatly circle and label each of these types of joints: ball and socket, hinge, pivot, gliding, ellipsoid.
- 7** Highlight the growth plates by colouring the upper and lower parts of the long bones of the limbs.

Keys and kingdoms

Animals and plants are so numerous that people need to classify them into groups and then into smaller groups to be able to talk about them. The groups are written in order of size, with the largest group first: Kingdom, Phylum, Class, Order, Family, Genus and Species.

There are four main reasons for classifying into groups:

- » **Communication** – So I can say to you the word ‘marsupial’ and you will know I mean a furry, warm, pouched mammal.
- » **Conservation** – If we know how many there are of a group and we see the numbers declining, we can take action to make sure they survive into the future.
- » **Ordering** – This happens so we know which ones came first and we can find things, like all the books in a library or all the files on your computer.
- » **Relationships** – These show which organisms are related to each other. For example, the platypus and echidna are both monotreme mammals.

1 Complete the table using the terms in the box.

+ absorbs food from trees	+ jointed legs	+ egg-laying mammal	+ thin, tubular and muscular
+ oyster	+ has a shell	+ spider	
+ mushroom	+ echidna	+ earthworm	

WHAT I SAY	EXAMPLE ORGANISM	WHAT YOU KNOW OR CAN LOOK UP	RELATED ORGANISM
monotreme	platypus		
fungus	toadstool		
mollusc	pipi shell		
worm	roundworm		
arthropod	ant		

Keys

A key is a convenient tool for finding out the name of an organism and what it is related to. At each stage in the key, you are given a choice of two groups. Choose one group at each stage and proceed through the key by following the instructions or answers on the right side.

2 Use the two-choice (dichotomous) key at the right to answer these questions.

a Who owns the blue car?

b Who drives the red four-wheel drive wagon?

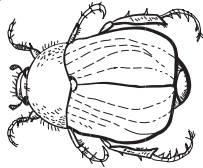
3 Referring to the key at the right, how many cars are there altogether?

KEY			
1	a	Red	Go to 2.
	b	Not red	Mrs Lee
2	a	Sports car	Mrs Miles
	b	Not sports car	Go to 3.
3	a	Two-wheel drive	Mr Silvestri
	b	Four-wheel drive	Mr Nguyen

- 4** **a** Use the pictures below to match the common name of the insect to its scientific order by working through the key at the right.
- b** Write each common name in the space on the key.
- c** Write each scientific order below the correct insect picture.

KEY TO SOME INSECT ORDERS			
1	a	Insect has wings, may be under a cover	Go to 2.
	b	Wings absent	Hymenoptera, _____
2	a	One pair of wings	Diptera, _____
	b	Two pairs of wings	Go to 3.
3	a	The two pairs of wings are different, the first pair is thicker than the second pair	Go to 4.
	b	Two similar pairs of wings	Go to 5.
4	a	First pair of wings hard and shell-like	Coleoptera _____
	b	First pair of wings leathery, sucking mouth parts	Hemiptera _____
5	a	Wings covered with scales	Lepidoptera _____
	b	Wings clear and membrane-like	Go to 6.
6	a	Wings with few veins	Hymenoptera, _____
	b	Wings with many veins	Odonta, _____

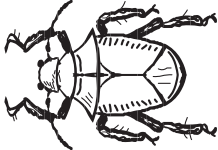
beetle



- two different pairs of wings
- first wings shell-like

Scientific order:

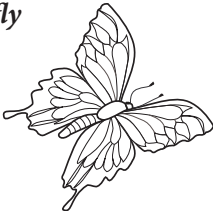
squash bug



- two different pairs of wings
- sucking mouth parts

Scientific order:

butterfly



- two similar pairs of wings
- scales

Scientific order:

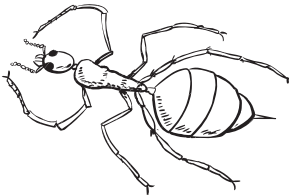
wasp



- two similar pairs of wings
- few veins on wings

Scientific order:

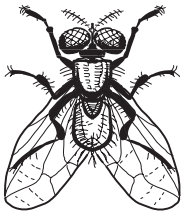
ant



- no wings

Scientific order:

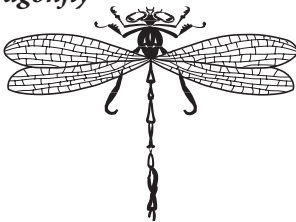
fly



- one pair of wings

Scientific order:

dragonfly



- two similar pairs of wings
- many veins on wings

Scientific order:

- 5** Design a two-choice (dichotomous) key to separate these presents by their outside wrappings. Use the minimum number of words. When you finish the key, each name should be separated somewhere along the key.



Jenni



Martina



Phuong



Ben



Rafael



Quan

Mixtures

Mixtures are all around us. Air is a mixture, and so are soils and the water in the ocean. There are mixtures of elements and there are mixtures of elements and compounds. These drawings represent mixtures and compounds:

Every sweet is the same
COMPOUND



Many different sweets
MIXTURE

- 1 Write the substances that each mixture is made from. Choose from these words, which you may use more than once: salt, sand, vegetables, nitrogen, oxygen, clay, water, carbon dioxide, sugar.

MIXTURE	SUBSTANCES
mud	
air	
sweet drink	
sea water	
soup	

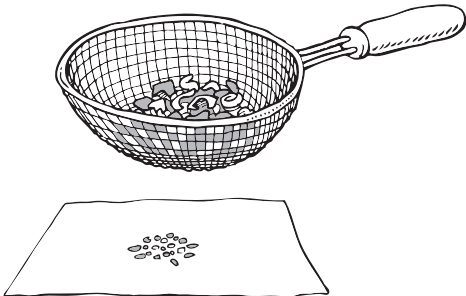
Water dissolves many substances and so is called the ‘universal solvent’. People often want to separate mixtures so they can use the substances in different ways or so the end product is more acceptable to the consumer. No one wants to drink water that has little bits of soil in the bottom or dead leaves floating on the top.

Methods of separating mixtures

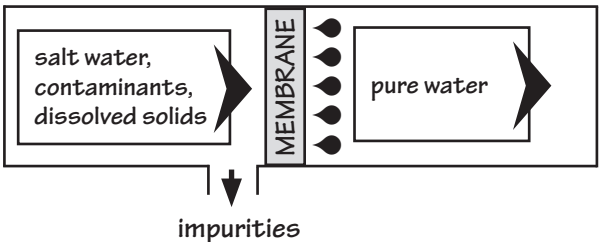
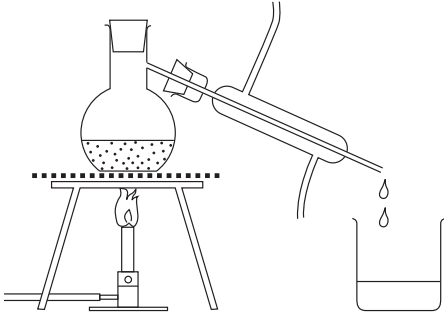


Filtration

Sieving



Distillation



Reverse osmosis

- 2** Write the correct method of separation – filtration, sieving, distillation or reverse osmosis – beside each description below.
- a** Water pressure is applied to the saltwater side of a semi-permeable membrane, forcing fresh water through and leaving the salt behind.

- b** Impure water is boiled. The steam is collected and cooled down in a long tube. Pure water runs into a container.

- c** Water runs through a substance that holds back larger molecules. _____
- d** Water runs through a substance that holds back smaller molecules. _____
- 3** Write a definition for each method of separation, and include a brief description of a scientific situation in which that method would be used.
- a** filtration _____

- b** sieving _____

- c** distillation _____

- d** reverse osmosis _____

Practical applications

- 4** People often want to separate the following substances. For each one, describe the best method of separation and justify your answer.
- a** Coffee grounds from coffee liquid _____

- b** Sand from water _____

- c** Salt from water (Remember, salt is completely water soluble) _____

- d** Lumps from sugar _____

- e** Sultanas out of flour _____

Natural resources

A resource is a source of materials that we can use. A resources boom is an upsurge in the economic activity and general prosperity of a country or region as a result of the widespread development of mineral resources like coal or uranium. Western Australia is experiencing a huge resources boom in iron at the moment. In the 1850s there was the goldrush in Victoria when 370 000 immigrants arrived in Australia and the economy of the nation boomed.

We can get natural resources from the ground, the air, the water or plants. All the minerals and ores in the ground are natural. Humans make many products from these natural resources.

- 1
- Beside each resource, write the product that can be made from it. Some resources may have two products.
PRODUCTS: glass, plaster of Paris, steel, paper, window frames, cement, bricks, water pipes, plastics, glues, jewellery, batteries, dinghies, integrated circuits, yellow cake.

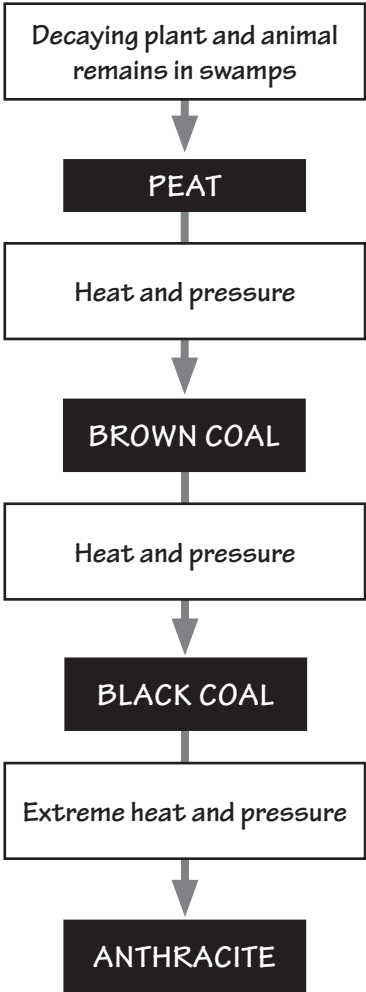
RESOURCES	PRODUCTS
sand	
clay	
petroleum	
gold	
copper	
trees	
silicon	

RESOURCES	PRODUCTS
iron	
gypsum	
heavy metals	
aluminium	
gums or plant resins	
limestone and clay	
uranium	

The formation of coal

A flow chart is a useful method of presenting data when observations of a sequence of events are made. Refer to the flow chart at the right to answer the following questions.

- 2
- Identify the material that is converted to peat.
-
- 3
- Describe how brown coal is formed from peat.
-
- 4
- Describe the texture of black coal.
-
- 5
- List the final product and describe its appearance.
-
-
- 6
- Explain why a flow chart is a more effective way of presenting this information than a paragraph of writing.
-
-



The Sun

7 Use the following word bank to help fill in the answers in the passage below: coal, oil, petrol, fossils, energy, Sun, process, generators, millions.

The original source of energy for Earth is the _____. The _____ comes to Earth in the form of electromagnetic waves. Plants convert sunlight into food. Layers of buried plant material can become coal. We can still see some of the leaves of these plants today because they have become _____.

Oil is formed from large amounts of small marine organisms that have been buried. Most coal and _____ was formed _____ of years ago. We burn oil in our cars as _____ and in planes as kerosene. Industry burns _____ in huge _____ to produce electricity. Coal is also used as a source of heat or as part of the chemical _____ to produce metals.

8 Extract information from the passage above to complete the following.

a Identify the original source of most of the energy on Earth. _____

b Explain why oil and coal are called 'fossil fuels'.

c Use three stages to explain how the energy ends up in the coal.

Stage 1: _____

Stage 2: _____

Stage 3: _____

d Describe two uses of oil and two uses of coal.

e Explain how the burning of fossil fuels has contributed to the Greenhouse Effect.

Australia's natural resources boom

9 Order the information at the right by listing the three largest contributors to Australia's revenue.

a Top revenue raiser _____

b Second revenue raiser _____

c Third revenue raiser _____

10 These figures are hard to compare because they are not measuring the same factors. Recommend one specific way of making these figures easier to analyse.

+ A\$ 22 billion – the value of black coal exports in 2005

+ 29% – Australia's share of total global coal exports, the world's highest

+ 1 143 000 – the number of tonnes of known recoverable uranium. It represents a quarter of the world's total

+ A\$ 6.9 billion – liquefied natural gas (LNG) exports are expected to rise 51% to this figure in 2006–07

+ A\$ 130 000 – expected salary of a newly qualified mining engineer, compared with A\$ 29 000 for a pharmacy graduate

+ A\$ 17.4 billion – in 2006, iron ore exports grew by nearly 30% to this amount

+ 305.2 million – the number of tonnes of iron ore that will be produced in 2006–07

Organs and organ systems

Your body has a variety of organs, ranging from your brain to your stomach and skin.

A **system** is a collection of organs, blood vessels and tubes that work together.

- 1
- Write each organ beside the correct system: brain, stomach, large intestines, lungs, liver, kidneys, bronchi, urinary bladder, small intestines, heart, pelvis, ovaries, testes, uterus, pancreas, skull, vertebrae, spinal cord.

THE SYSTEMS	ORGANS IN THE SYSTEM
digestive	
nervous	
reproductive	
excretory	
respiratory	
circulatory	
skeletal	

The function of each system

All our systems work together under the control of the brain to keep us in good health. The scenario below shows what can happen when something goes wrong.

It is a very hot day in the middle of summer and Jenny and Sean have decided to paint their house. They collect all their gear, put on old shirts and shorts and head outside to paint. When they have been working for two hours, Jenny begins to feel faint. Sean checks her pulse, finding it to be weak and fast, but her skin is sweaty, and cool.

- 2
- Complete the following questions using the ideas bank below.

+ heat exhaustion	+ not able to lose	+ not enough	+ wear hat	+ brain
+ shade	enough heat	feedback to brain	+ don't work in the	+ skin
+ wet cloths	+ drink water	+ sit in shade	middle of hot days	+ kidneys

- a
- Describe what has happened to Jenny, and suggest first aid measures that could return her body to its normal functioning.

- b
- Identify the organs that have contributed to Jenny's condition.

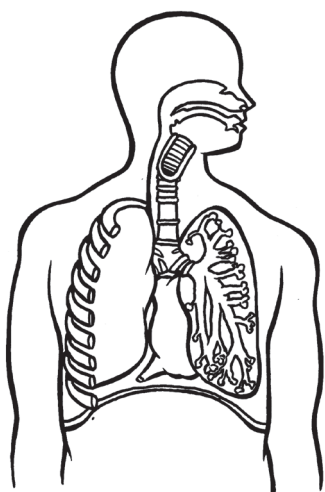
- c
- Why didn't Jenny realise that she was getting heat exhaustion?

- d
- What should Jenny do after every hour of work to prevent this from happening again?

Human body systems

3 Label the diagrams using the words listed under each one.

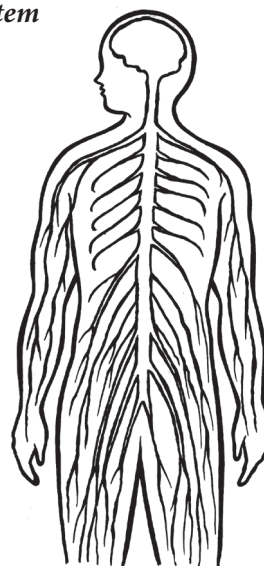
Respiratory system



Labels: lungs, heart, bronchi, trachea, diaphragm

Function:

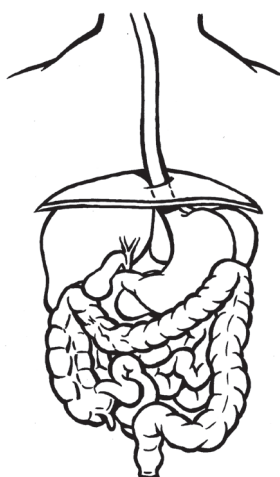
Nervous system



Labels: brain, spinal cord, nerves

Function:

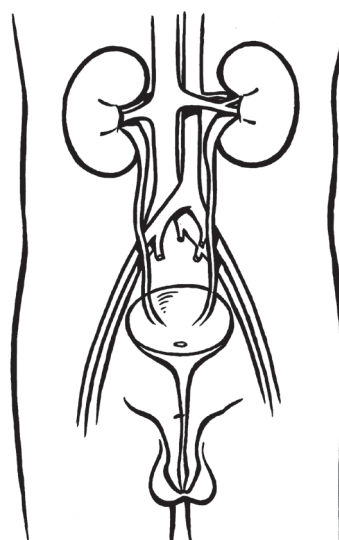
Digestive system



Labels: stomach, liver, large intestine, small intestine, gall bladder

Function:

Excretory system



Labels: kidney, bladder, ureters, urethra, blood vessels

Function:

4 Listed below are four jobs or functions carried out by these systems. Write the correct function under each diagram above.

- Removes waste urea from the blood and maintains the correct water balance
- Changes food into water-soluble form so the blood can carry it around the body
- Adds oxygen to the blood so it can be carried to the cells
- Manages the interaction of all the systems and all the movement of the body

Particle model of matter

Everything you do each day relies on aspects of the environment staying constant. For example, furniture and floors keep their shape, water flows from pipes or drops from clouds and air moves easily past your body. These are the three states of matter: **solid, liquid and gas.**

Particles and space

Alinga and Adrian took a cup full of hot water and sprinkled a teaspoon of sugar on it. The sugar dissolved and the cup did not overflow. They added more teaspoons of sugar by sprinkling and saw that 14 teaspoons could be added before the cup overflowed. They kept the water hot by standing the cup in a saucepan of hot water. How did all that sugar fit into the cup when it was already full of water?

1 Present your best explanation or hypothesis as a paragraph. These words may be helpful: spaces, holes, sugar, water, moving, squeeze, heat, faster, dissolves.

Introduction_____

Middle_____

Conclusion_____

2 Solve these problems.

a Salt molecules are smaller than sugar molecules. Do you think Alinga and Adrian could dissolve more or less teaspoons of salt in their cup of water than they could of sugar? Explain your answer.

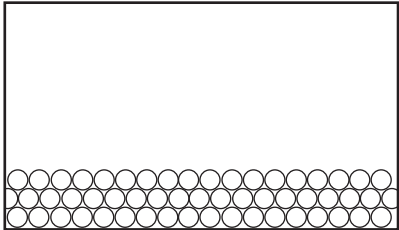
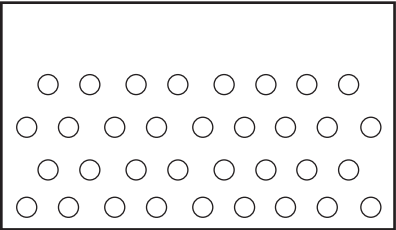
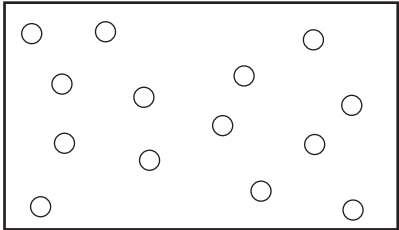
b Suggest why Alinga and Adrian sprinkled the sugar instead of pouring it on the water in the cup.

Once scientists have observed that air can be blown into a balloon or that tyres can gain very high air pressure they make an hypothesis to explain their observations. These and other observations were brought together and called the Particle Theory of Matter.

All matter is made of tiny particles that can be pushed together until the substance becomes solid or liquid. Water becomes ice. Gases can become liquid, like liquid oxygen. Gas, liquid and solid are called the three states of matter.

3 Write each label under the correct diagram on the next page.

+ particles moving rapidly	+ particles stay in the bottom of the container	+ particles often collide
+ particles moving moderately		+ particles jostle each other
+ particles just vibrating	+ particles collide constantly	+ particles fill the container
+ particles stay in place		

Solid	Liquid	Gas
		

Change of state of solids, liquids and gases

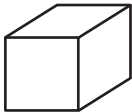
Gases can expand and contract. When a gas is compressed, its pressure rises and eventually it can form a liquid. Liquids can evaporate or solidify. Most liquids decrease in volume when they condense. Solids like iron can be melted, often great heat is required to do this.

4 Draw a cycle diagram in the space below to show how water can change states.

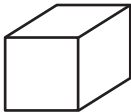
Density

Density is the number of particles in a certain volume. It is measured in grams of mass per cubic centimetre of volume. Water is taken as the basic unit and its density is 1 gram per 1 cm³.

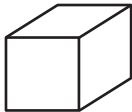
5 Given that 10 particles represent 1 gram and 20 particles represent 2 grams, draw the correct number of particles in each cube. Use one dot to show one particle.



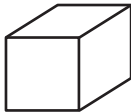
Water
1 g/cm³



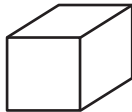
Aluminium
3 g/cm³



Iron
8 g/cm³



Lead
11 g/cm³



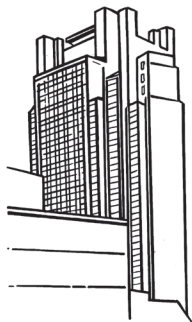
Gold
19 g/cm³

6 Describe what you found out from drawing the particles in the previous question.

Qualitative work in science

In Art, History, English or Science, you may be asked to describe the characteristics of an object or person. Your audience wants to know the special qualities or properties of that object. This is a **qualitative analysis**. **Quantitative** work involves numbers. For example, you could measure the height and width of an object, count the number of branches of a tree or calculate the quantity of concrete in a building.

- 1** Under each illustration, list the features or properties of the object.



Apartment block



Eucalypt



Pine tree

- | | | |
|---------|-------------|------------------------------|
| • _____ | • one trunk | • narrow branches at the top |
| • _____ | • _____ | • _____ |
| • _____ | • _____ | • _____ |

- 2** Now use your lists from question 1 to write a qualitative analysis of each object. A reader who cannot see the illustration should still understand what is being described.

a The apartment block is _____

b The eucalypt is _____

c The pine tree is _____

The properties of substances

In the mineral world, properties can be described qualitatively by their lustre, hardness, colour, magnetism and density. Qualitative chemical analysis of compounds tells us the chemicals that are present, but not how much of a substance is present. In Physics, we can relate speed, distance and time to each other without using measurements to gain a qualitative explanation of what is happening.

- 3** Complete these sentences to explain qualitatively each phenomenon. Use terms from the information above to help you.

- Qualitative mineralogy – Diamonds are valuable because they have a bright _____ and they are rare. They can be used to grind away glass and concrete because _____.
- Qualitative chemistry – When common salt, sodium chloride NaCl, is qualitatively analysed it is found to be made up of the elements _____ and _____.

- 4** Qualitative physics

- a** A large truck moving fast downhill will do great damage to a brick wall at the bottom of the hill. A small car will do less damage to the brick wall. In your workbook, relate mass, speed and time in a qualitative explanation of these forces.
- b** Invent a bicycle ride that will give people a thrill without hurting anyone. The bikes may be attached to tracks for safety. What qualities will you vary? In your workbook, write a qualitative explanation of your invention.

Respiration

There are two different meanings for respiration. In sport and first aid, respiration refers to getting the air in and out of your lungs in the respiratory system. In biology, it refers to respiration in the cells of your body. Here the oxygen is combined with glucose to produce energy for all your body processes.

Glucose + oxygen → carbon dioxide + water + energy



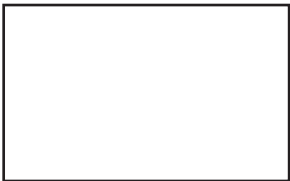
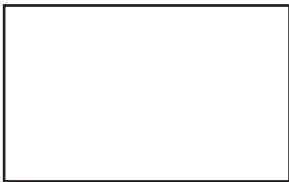
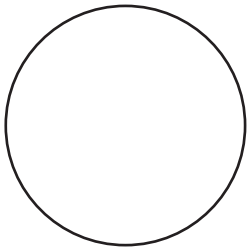
People eat a variety of foods, which are digested or turned into water-soluble form, and passed into the blood stream which surrounds the digestive system. Blood takes the soluble food to cells.

Blood also collects oxygen from your lungs and takes it to your cells. In the cells, the food and oxygen combine to release energy and the waste products carbon dioxide and water.

Respiration flow chart

1 A flow chart to show the stages of respiration has been started in the space below. The different shapes in the flow chart have different meanings, which are shown on the next page. Write the terms below in the shapes to complete the flow chart, and draw arrows to link the shapes. One term and one arrow have been done for you.

+ food enters mouth	+ cell respire	+ oxygen enters lungs	+ oxygen enters lungs
+ transport of soluble food via blood	+ energy and wastes produced	+ transport of oxygen via blood	+ digestion



Flow chart key

input or processing
takes place

decision
point

input or
output

starting,
stopping,
control
point

Photosynthesis

Plants can photosynthesise, or make their own food. They use carbon dioxide from the air, water from the soil and energy from the Sun to make glucose and oxygen.

All animal cells respire and plant cells also respire. They all need energy.

2 Contrast respiration and photosynthesis.

Respiration	Photosynthesis

3 In your opinion, which process is more important – respiration or photosynthesis? Justify your choice by explaining how animals could live without plants or how plants could live without animals.

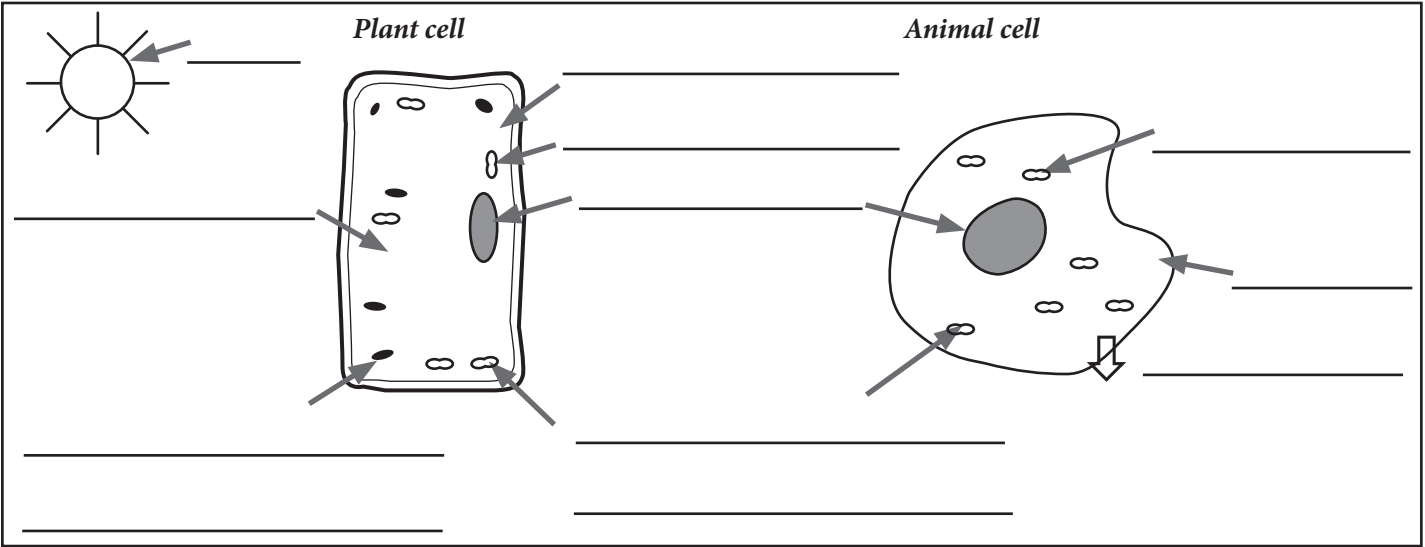
My opinion _____

Justification _____

The energy produced

Different cells use energy for different functions. Muscle cells use it to contract and relax, skin cells use it to grow new skin and bone cells use it to lay down bone. All cells use energy to make new protein and to reproduce. When a cell has no energy, it dies.

4 Use these terms to label the diagram: Sun, carbon dioxide, water, chloroplast – site of photosynthesis, mitochondria – site of respiration, nucleus, oxygen, glucose, energy out.



Solar system

Our solar system is made up of a medium-sized star called the Sun, eight planets, an asteroid belt and a comet belt. The four planets closest to the Sun are made from rock. The other four planets are called gas giants and are made up of various gases, but mainly hydrogen, helium, methane and ammonia. Mercury has no atmosphere while Venus and Mars have atmospheres made up of carbon dioxide. Earth's atmosphere is made up of nitrogen, oxygen and carbon dioxide and other gases.

Mercury and Venus do not have moons and Earth, Mars, Jupiter, Saturn, Neptune and Uranus do have moons. The moons are called satellites because they orbit their planet. All moons rotate. The Sun rotates every 27 days. Each planet rotates at a different speed on its own axis. All of the planets orbit the Sun in the same plane and their orbits are elliptical. The distance from the Sun to the Earth is 150 million kilometres, which equals 1 astronomical unit (AU).

1 Complete the table using the information above.

	Rock or gas?	Atmosphere	Moons?	Distance from the Sun (AU)
MERCURY				0.4
VENUS	Rock			0.7
EARTH		Nitrogen, oxygen, carbon dioxide, other gases		1
MARS				1.5
JUPITER				5
SATURN				9.5
URANUS			yes	19
NEPTUNE				30

2 In your workbook, draw the distance of each planet from the Sun. Place the Sun at the bottom of the page. Use a scale of 1 AU = 1 cm, and consider carefully how to orientate your diagram so that you can include Neptune.

3 Calculate the distance from the Sun in kilometres for each planet. Remember, 150 million km = 1 AU.

PLANET	KILOMETRES
Mercury	
Venus	
Earth	
Mars	
Jupiter	
Saturn	
Uranus	
Neptune	

Day and night

4 Use these words to complete the passage below: reflected, spins, sunsets, south, daylight, night, imaginary, east, hours, phases, 24, bright.

The Earth _____ on its axis once every _____ hours, giving us day and night. The axis is an _____ line which joins the north and _____ magnetic poles. When one side of the Earth faces towards the Sun, that side has _____. The side facing away from the Sun experiences _____.

The Earth turns from west to east, which causes the Sun to rise in the _____. If you live at the latitudes closest to the Equator, you will experience short _____. Closer to the poles, sunset can take many _____.

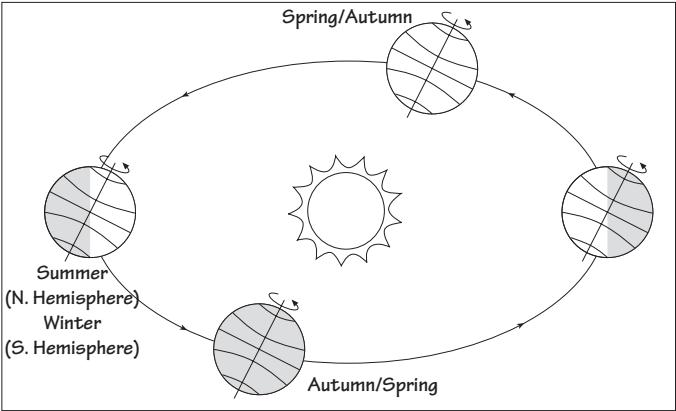
Our Moon reflects the Sun's light. This means that sometimes the Moon is a very _____ object in the sky and sometimes it is completely dark. As the Moon revolves around the Earth we see different _____, depending on how much light is _____ from the Moon back to Earth.

The seasons

The Earth’s tilt is 23.5° and this tilt causes the seasons – summer, autumn, winter and spring.

If the Earth did not tilt as it circled the Sun, the amount of daylight would be approximately 12 hours both above and below the Equator. The seasons depend on how many hours of sunshine we receive each day.

In summer in Australia, we receive about 14 hours of daylight, which heats the land and causes high temperatures. At the same time, the northern hemisphere receives about 10 hours of light per day and they experience winter. At the Equator, there is little variation in the lengths of the days.



5 Using the information above, complete the following table for today. Choose a specific place in each hemisphere.

	SOUTHERN HEMISPHERE	NORTHERN HEMISPHERE
Chosen place		
Month of the year		
Season of the year		
Average hours of daylight		
Average hours of darkness		
Average temperature		

6 If Earth had a tilt of 46°, how do you think this would affect each hemisphere?

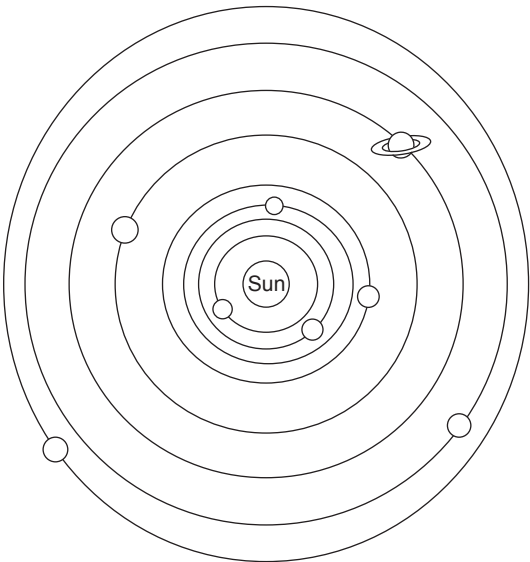
Southern hemisphere _____

Northern hemisphere _____

Motions of the planets

Each planet has its own orbital speed around the Sun. The orbits of the four inner planets are measured in Earth days, while the gas giant orbits are measured in Earth years.

	MERCURY	VENUS	EARTH	MARS	JUPITER	SATURN	URANUS	NEPTUNE
Earth days	88	225	365	687				
Earth years					12	29	84	165



7 Label each planet on the diagram of the solar system. Then draw the positions of the planets 88 days after the starting positions. Note that the diagram shows the correct order of planets, but is not drawn to scale.

8 How does Mercury appear to move, relative to Earth?

Technology

Technology is a broad concept that deals with our use and knowledge of tools and crafts, and how they affect our ability to control and adapt to the environment. Most technological advances emerge from the fields of science and engineering.

- 1 What do you consider to be the greatest technological advance in the last 100 years?_____
- 2 Evaluate your answer to question 1 by describing the technology, outlining how it makes life easier or more convenient and comparing the way people lived before and after this technology was developed.
- Name of technology_____

• Description_____

• Outline_____

• Comparison of before and after_____

Devices

Devices are designed to make a job easier or to enable you to do something that is not otherwise possible. A device can be as simple as a can opener or as complex as a generator in a power plant that produces electricity for a city.

- 3 Complete the table by listing all the devices that you see or use in one week, for each category.

SIMPLE TOOLS	TOOLS WITH MOVING PARTS	DEVICES USING POWER
Knife and fork		

Applications of technology

- 4 Imagine that you want to move some soil to make a skateboard park with a variety of ramps. List the appropriate device for moving each quantity of soil beside the job to be completed: spade, small spade or trowel, front end loader, motorised backhoe.

JOB	DEVICE
Collect sample of soil for testing	
Move soil to create wall shapes before concreting	
Remove debris from site	
Planting final site with shrubs	

Levers

A simple device is often a **lever arrangement**. This means there is a place to hold the object and also apply the effort; a balancing point (the fulcrum); and a place where the work is done (the load).

The further the fulcrum is from where the effort is applied, the more power you generate.

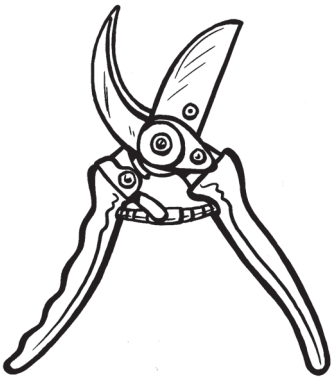
- 5 a Label the secateurs with these terms: effort, fulcrum, load.
- b Draw arrows on the secateurs to show the direction of movement while effort is being applied.

Force x distance = force x Distance

In the equation above, *Force* with a capital F means a large force and *force* with a small f means a small force. Similarly, *Distance* means a long distance and *distance* means a short distance.

- 6 a Use the above equation to write in the table the Force that will be applied at the cutting end of each object.

OBJECT	Force (kg)	distance (cm)	force (kg)	Distance (cm)
A		10	10	80
B		10	10	60
C		20	10	40
D		20	10	20



- b What kind of object is A? _____ What could you cut with it? _____
- c What kind of object is D? _____ What could you cut with it? _____

Friction

One of the biggest problems with technology is that most devices have moving parts which develop friction. As the moving parts rub together or rub against air or water, they use up some of the energy. This rubbing or friction slows us down, slows down cars, makes engines wear out and makes it difficult to send rockets into space. However, if there was no friction we would fall over constantly and cars would not stop!

- 7 Complete this table by listing the advantages and disadvantages of friction for each object or person. Then briefly describe how the friction can be overcome.

OBJECT / PERSON	DISADVANTAGE	ADVANTAGE	OVERCOMING FRICTION
car tyres		Grip on wet roads	
rockets			
ice skater			
car motor			
running person			Run on toes, wear tight-fitting clothes

A building project

- 8 Some of the blocks used to build the pyramids in Egypt weigh 2.5 tonnes. In your workbook, explain how the pyramids could have been built. Include a labelled diagram in your answer.

Useful terms					
+ ramp	+ rope	+ exterior	+ wheels	+ labourers	+ chocks
+ pulley	+ interior	+ rollers	+ lever	+ lubricant	+ raft

V ariables

When you want to find out the answer to a problem, you need to know exactly what causes the end result. You need to do an experiment! This provides reliable, first-hand data.

If your television is not working well, you change just one item at a time to see if that gives you the result you want. You would not change the aerial, the cable, the position and the remote at the same time, or you wouldn't know which change made it work better. All the items that can be changed are called variables.

Imagine that your teacher has set two assignments for your class. The first is to find out which shape of pasta cooks fastest and the second is to find out which brand of hair shampoo makes the most foam. To conduct fair experiments, you need to keep many items the same – that is, control your experiment.

1 Complete the following tables by writing 'keep the same' or 'vary' in the columns.

Pasta experiment

ITEMS	KEEP THE SAME OR VARY?
amount of water	
water temperature	
size of saucepan	
time intervals measured	
amount of pasta	
shape of pasta	

Shampoo experiment

ITEMS	KEEP THE SAME OR VARY?
amount of water	
water temperature	
size of container	
height of layer of bubbles	
amount of shampoo	
brand of shampoo	

The independent variable

This is the most critical part of the experiment. The independent variable is the item that 'I change' and it is the actual experiment.

The result depends on what is changed, and is called the dependent variable.

Tip!

A memory helper –
The I for Independent is
the item that I change.

Working with variables

2 Design experiments to find out the following. You only need to set out the variables.

a What brand of ice-block stays on its stick the longest?

- Independent variable – I will change this: _____
- Controlled variables – keep these the same

- Dependent variable – this is the result: _____



b How much fertiliser do plants need for optimum growth?

- Independent variable – I will change this: _____
- Controlled variables – keep these the same:

- Dependent variable – this is the result: _____



Waste from resource use

Waste management is a billion dollar industry. The industry leaders attend large conferences such as Environmental Management in Mining and Promoting Sustainable Practices to gain the latest knowledge about waste management. One key to managing waste is to find new profits in waste and in ways of restoring the environment. Here are two examples of changing a cost into a profit:

- » Turning areas with non-toxic waste into recreation and playing fields by smoothing them over, replanting and selling them to a consumer group.
- » Power stations that produce 13 million tonnes of fly ash, a mineral residue from burning coal, can sell it for road base instead of paying for landfill space.

1 Think of uses for these wastes that could earn money. Think big!

a Waste: Used tyres, equivalent of 30 million passenger vehicle tyres each year

Use _____

b Waste: Heat given off from large cooling towers

Use _____

c Waste: Used newsprint paper, 600 000 tonnes each year

Use _____

2 Pollution is a general word for contamination by unwanted substances. Write each source of contaminant beside the correct effect: agricultural lands, mining leachate, urban roadways, untreated sewage, smokestacks and exhaust pipes.

SOURCE OF CONTAMINANT	EFFECT
	Fertiliser runs into stream and excess algae grows
	Viral and bacterial pathogens cause disease
	Heavy toxic metals from moving engine parts poison aquatic fauna
	Heavy toxic metals move into groundwater in aquifer
	Sulfur dioxide spreads into air and forms acid rain

Strategies

We need strategies in place to balance human activities and needs with conserving and protecting the environment. Solving problems involves two stages: choosing a strategy for solving the problem and then implementing that strategy. Choosing is the more important part and the more difficult part.

3 Evaluate the strategy proposed to offset the following problem.

Planes are huge users of fossil fuel, yet more people want to fly each year. One airline is heavily investing in the development of the alternative fuel bioethanol, which is made from corn. Corn is a renewable resource as it can be grown every year.

- Positive points _____
- Negative points _____
- Overall judgement _____

Reduce, reuse, recycle!

4 Conduct an environmental assessment of your home or school, and compile a 10 point plan in your workbook for decreasing waste by reducing, reusing and recycling.

Xylem and phloem

The xylem and phloem tubes in plants are similar to the arteries and veins in animals, but there are some important differences.

- » Xylem – Carries water up to the leaf. As the water evaporates out of the leaf, more water comes up through the xylem tubes or vessels from the roots.
- » Phloem – Carries sugar down from the leaves to all parts of the plant.

Xylem and phloem tubes are called vascular tissue, and so the plants are called vascular plants. Photosynthesis in the chlorophyll of the leaf is also an important plant process.

1 Complete the table at the right using phrases from the information above.

How growth in plants happens

- » Leaves make the sugars for the fruit to grow. Leaves need sunlight, carbon dioxide and water to photosynthesise.
- » The water comes up to the leaves from the roots via the xylem tubes.
- » The CO₂ comes into the leaves through tiny pores called stomata.
- » The sunlight shines on the chlorophyll in the leaf, giving it energy for the photosynthesis reaction.
- » The sugars that are produced travel along the phloem tubes to the fruit and to the roots to give them energy. The roots absorb water continuously and send it into the xylem tubes to the leaves.

	FEATURES
Arteries and veins	• Carry oxygenated blood away from heart
	• Carry deoxygenated blood back to heart
	• Blood goes around the body and returns to the heart
Xylem and phloem	•
	•
	• Non-returning fluid flow

What a whopper!

Port Lincoln in South Australia can lay claim to Australia's largest pumpkin. Weighing in at 440 kilograms, Ken and Rosemary Holden grew the giant from Atlantic Giant pumpkin seeds.

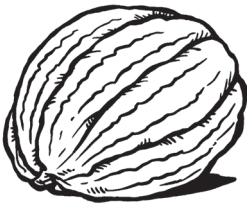
'We've put plenty of interesting things into the soil for the pumpkin. I dug a pit over half

a metre deep, five by three metres in area and filled it up with seaweed, pig manure, horse and cow manure and fish waste,' Ken Holden, Pumpkin Grower, said.

The big pumpkin will be turned into soup for the local hospital and meals on wheels.

April 2003 Permission R Holden.

Atlantic Giant pumpkin



2 You have decided to enter the giant pumpkin competition. How can you increase your chances of winning? Write a suggestion after each heading below. (One is done for you.)

Useful terms: prepare, compost, sow early, manure, green house, green crop, transplant, no insects, pollen, dig in, pH 6.5–6.8, other small pumpkins, excess stem growth, nitrogen, phosphorus, potassium, length, height, written records, water constantly, large growing variety, 8 hours

- Soil _____
- Seeds _____
- Pollinate _____
- Sunlight _____
- Water _____
- Reposition the pumpkin when it is small so that there is a right angle between the main stem and the pumpkin stalk, so the xylem and phloem have clear flow in their tubes.
- Remove _____
- Fertilise _____
- Measure _____

Yeast and yoghurt

Yeast is a friendly micro-organism that produces the gas carbon dioxide to give us bubbles in beer and ‘holes’ in our bread. Yeast also provides the characteristic ‘yeasty’ flavour of beer, champagne and bread. Even vegemite owes its flavour to yeast.

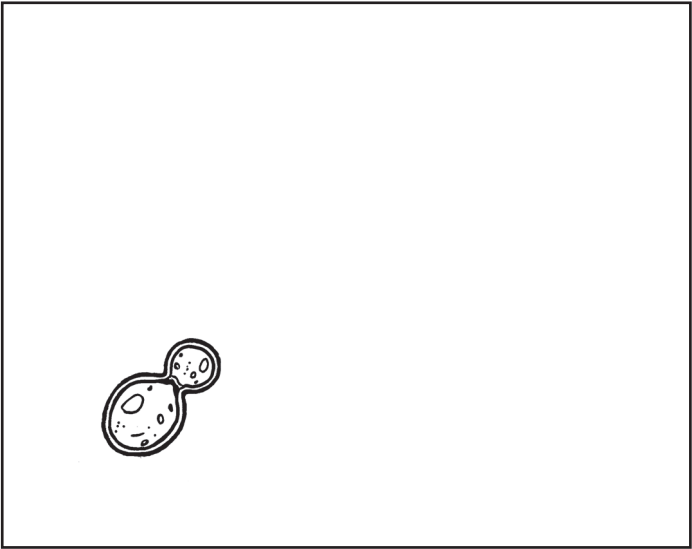
Reproduction in unicellular organisms takes place by cell division. Yeast is a single cell fungus. It reproduces asexually by budding. A small bud emerges from the surface of the parent cell and enlarges until it is almost the size of the parent.

- 1** In the box at the right, draw three more yeast cells to show them budding. Use dots to represent the cytoplasm.

Yeast respires with air or aerobically to produce carbon dioxide and water.

Yeast can respire without air or anaerobically to make carbon dioxide and alcohol.

When the alcohol reaches a specific concentration, it kills the yeast and stops it working. There are different strains of yeast for different purposes. The limit for beer yeast is 5% and for champagne it is 18%.

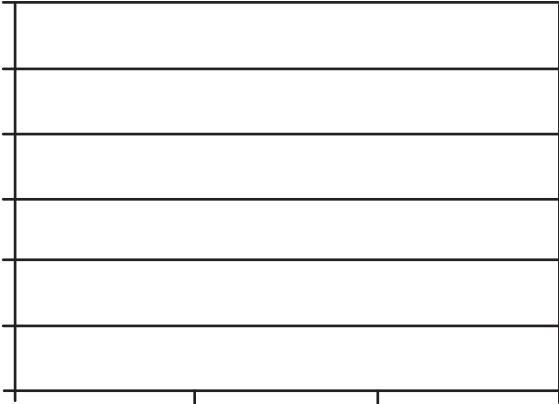


Graphing yeast reproduction

- 2** Lucine and Chen work in a scientific laboratory. They have been asked to grow yeast by feeding it warm dilute sugar solution. The table below shows their results of counting the cells under the microscope. Graph this data by labelling the axes, plotting the points and drawing a line of best fit.

TIME INTERVAL	YEAST CELLS
1	10
2	20
3	40
4	80
5	160
6	320
7	640
8	1 280
9	2 560
10	5 120
11	10 240

Yeast cell growth



- 3** Predict how many cells there will be in the next four time intervals.

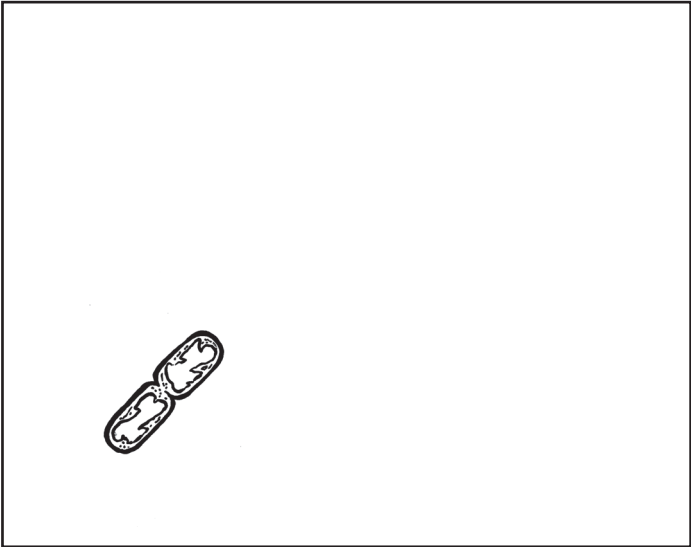
- a** time interval 12 _____
- b** time interval 13 _____
- c** time interval 14 _____
- d** time interval 15 _____

- 4 a** Recall other types of data that increase in a similar way.
- b** Describe what can happen if populations breed exponentially like this.

Yoghurt

Yoghurt is a fermented milk product, which was apparently brought to Turkey by the Mongols millennia ago. Adding a ‘starter’ of active yoghurt which contains a mixed culture of lactobacillus acidophilus and other bacteria produces lactic acid during the fermentation of lactose. The lactic acid lowers the pH, makes it tart, causes the milk protein to thicken and acts as a preservative, as pathogenic bacteria cannot grow in acid conditions.

Asexual reproduction involves only one individual or parent. The offspring generated by asexual reproduction are exact duplicates of the parent. Binary fission is the process by which a bacterium splits into two cells. Each cell gets an exact copy of the parent cell’s genetic material.



- 5** In the box at the right, draw three more cells doubling by binary fission.
- 6** Each of these products or diseases is caused by bacteria. Identify each one as either beneficial or harmful, and include a brief reason for each one.

PRODUCT OR DISEASE	BENEFICIAL OR HARMFUL? WHY?
anthrax	
soil enrichment with legumes	
cheese	
bacteria in dental plaque	
diphtheria	
yoghurt	
pneumonia	
bacteria in gut	
whooping cough	

- 7** Choose one harmful product or disease from question 6. Find out the history of scientific research of the associated bacteria.

Zoology

When you go to the zoo, you see lots of different animals. Zoology is the study of animals. In the zoo you can see large animals from each of the vertebrate groups. Invertebrate animals are often in places like butterfly farms, rather than zoos.

- 1 Write each animals beside the correct group: shark, zebra, parrot, bluetongue lizard, marsh frog, perch, platypus, plover, axolotl, crocodile, barramundi, turtle, corroboree frog, kookaburra, quokka, whiting.

GROUP	ANIMALS
Fish	
Amphibian	
Reptile	
Bird	
Mammal	

Vertebrate zoology

Vertebrates are sorted into groups depending on their anatomy, which includes their bone structure and their internal organs. Grouping vertebrates can assist in housing the animals, as those in the same group often have the same needs.

Reptiles need warmth in order to move around and to breed. They are **ectothermic**, which means that their temperature is the same as the surrounding environment. Different reptiles eat different foods such as insects or small mammals. They lay eggs and have scales. Birds and mammals are **endothermic** and can maintain a constant body temperature.

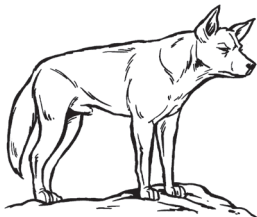
- 2 List four features of each group. One has been done for you.

FISH	
AMPHIBIAN	Ectothermic, moist skin, lay moist eggs, gills when young and lungs as adult
REPTILE	
BIRD	
MAMMAL	

Types of mammals

Australian zoos usually have the three types of mammal displayed: monotremes, marsupials and placentals.

- 3 Write each term below the correct mammal type. Some terms will be used more than once.
Terms: pouch, placenta, lays eggs, endothermic (warm blood), fur, suckles young.



Monotremes

Marsupials

Placentals

Variety

There is an enormous variety of animals in the world. Each group that breeds within its own group to produce fertile offspring is called a **species**. There are several million species of plant and animals on Earth. Each continent or land area has some species that are unique to that place, like the platypus is unique to Australia.

4 Complete the table.

UNIQUE ANIMAL	DESCRIPTION OF ANIMAL	CONTINENT
Tiger		
Llama		
Giraffe		
Bilby		
British hedgehog		
Bison		

Seventy percent of the world’s species mostly occur in Australia, Brazil, China, Columbia, Ecuador, India, Indonesia, Kenya, Madagascar, Mexico, Peru and Zaire. Tropical rainforests are believed to contain more than half of the total number of species on Earth. Bats account for a large number of mammal species in the tropics.

5 Suggest two possible reasons why there are more animals and plants in the tropical rainforests than in the places at higher latitudes.

a _____

b _____

A modern zoo

6 Imagine you are in charge of the zoo. Propose an arrangement for an enclosure for an animal of your choice. Consider the following ideas in your proposal. A drawing may show your ideas more clearly.

- Animal species: _____
- Vertebrate group: _____
- If a mammal, specify class of mammal _____
- Food and water requirements _____
- Size and description of enclosure _____

- Social needs and number of animals in the enclosure _____

- Climate needed _____
- Weather protection, day and night _____
- Ways to keep the animal active, alert and happy _____

Selected answers

Adaptations

- 1 Examples could include hare, elephant, bilby, fox
- 3 Examples could include strong jaws for eating plants; tiny feelers near the mouth for tasting the plants; strong back legs for jumping; large back wings for flying.
- 4 possum – flexible tail to climb trees; banksia tree – thick seed cones to resist fire; wombat – strong front claws to dig; sundew – sticky hairs to catch insects; kookaburra – pointed long beak to grab lizards.

Botany

- 2 **a** moss, small leaves, spore capsules
b conifer, pine needles, seeds in cones
c algae, no stems, no leaves
d fern, fronds, spores
e flowering plant, flowers, seeds in fruit
f flowering plant, stem, seeds in fruit
g moss, small leaves, no stems
h algae, no stems, no leaves
- 3 carbon dioxide + water → glucose + oxygen
- 4 the air
- 5 into the air
- 6 Encourage students to consider grains and pulses as well as fruit and vegetables

Cells

- 1 common, organelles, chloroplasts, nucleus, moving, holds, through, jelly, covering, drying, structures, space
- 2 cell membrane – holds the cell together; cell wall – helps the plant keep its shape and stops it drying out too quickly; cytoplasm – allows all the fluids and nutrients to have a base; nucleus – controls the cell's jobs like growing, moving, reproducing and eating; chloroplast – makes food; vacuole – storage

Data gathering

- 3 (In table order) thermosensor, heart rate sensor, pulse oximeter, thermoanemometer, expendable primary element with data transmitting assembly, thermometer, pH meter, flow meter, chloride probe, nitrogen probe
- 4 **a** Doppler radar, Bureau of Meteorology.
b waves, from which speed of approaching storms can be calculated
c Adelaide, Melbourne, Sydney, Brisbane
d Could be used to measure the speed of any object, e.g. speeding cars, aeroplanes.
e The types of waves shown on the radar would indicate that hail is approaching, not rain. The direction in which the waves on the radar are bunching up indicates the direction in which the storm is going. The degree of bunching indicates the speed of the storm's approach.

Ecosystems

- 1 Biotic: grass, bark, leaves, wombat, caterpillar
Abiotic: sunlight, temperature, chemicals, water, minerals, oxygen
- 4 Producer: bottlebrush, algae, grass
Consumer: wedge-tail eagle, cockroach, frog, witchetty grub
Decomposer: fungus, bacteria
- 6 plants (producer), caterpillar (consumer 1), small bird (consumer 2), eagle (consumer 3), wildcat (consumer 4)
- 8 **a** glucose, oxygen
b oxygen, carbon dioxide, energy

Forces

- 3 Everything inside will float, it is weightless because Earth's gravity is no longer acting on it.

5

Force	Contact force	Non-contact force
magnetism	not applicable	picking up nails
electrostatics	not applicable	picking up paper
gravity	not applicable	falling to the ground
buoyancy	ship floating on water	not applicable
air resistance	rocket heating up as it travels through air	not applicable

- 6 **a** Gravity while falling, opening a parachute to increase air resistance and reduce speed of falling
b Use a strong magnet, as magnetism is a non-contact force and will collect the paperclips you can't reach.
c Use electrostatic force by rubbing a cloth against an insulated rod, then using the rod to pick up the balls
d Buoyancy of the ship from the water, engines to move the ship across the water.

Graphing

- 1 **a** Height gradually increases with age, between the ages of 7 and 14 years.
b The graph – the direction of the graph shows you the change each year, without having to calculate the difference between the years.

- 2 No title, no units on the axes

3

Dependent axis	Independent axis
The girls' heights change	as they grow older
The boys' running speeds vary	as the years pass
The amount of petrol the cars use	the engine capacity
The lung capacity of teenagers	their height

Hydrosphere

- 1 potable, energy, faster, cycle, surface, gas, evaporation, droplets, temperature, precipitation, snow
- 3 gravitational, Moon, water, same, side, right, six, hydrosphere.
- 4 A – neap tides; B – spring tides; C – neap tides; D – spring tides

Investigations

- 1 **a** first-hand investigation
b second-hand investigation
c first-hand investigation
d second-hand investigation

Joints

- 1

Name of joint	Place in body	Type of movement
ball and socket	shoulder, hip	rotation
hinge	elbow, ankle	backwards and forwards
pivot	atlas, axis, top of neck	pivot in most directions
gliding	foot	glide on one plane
ellipsoid	wrist	two directions
- 3 long, adolescents, plate, bone, complete, weakest, connect, damage, fractures, boys, mature, solid

Keys and kingdoms

- 1 (In table order) egg-laying mammal – echidna; absorbs food from trees – mushroom; has a shell – oyster; thin, tubular and muscular – earthworm; jointed legs – spider.
- 2 **a** Mrs Lee
b Mr Nguyen
- 3 four
- 4 1b Hymenoptera, ant
2a Diptera, fly
4a Coleoptera, beetle
4b Hemiptera, squash bug
5a Lepidoptera, butterfly
6a Hymenoptera, wasp
6b Odonta, dragonfly

Mixtures

- 1 (Note other answers are possible)
mud: sand, nitrogen, oxygen, clay, water
air: nitrogen, oxygen, carbon dioxide
sweet drink: water, carbon dioxide, sugar
sea water: salt, water
soup: salt, vegetables, water
- 2 **a** reverse osmosis
b distillation

c sieving

d filtration

- 3 **a** Filtration: A method of separating solid particles from a liquid by pouring the mixture through filter paper in a funnel.
b Sieving: A sieve is a tool with mesh at the bottom which is used to separate small solid particles from larger ones. The small particles pass through the mesh and the larger ones do not.
c Distillation: The process of separating the components of a liquid mixture by boiling the liquid and then recondensing the vapour.
d Reverse osmosis: A process that separates impurities from water by forcing it through filters under great pressure.

Natural resources

- 1 sand – glass; iron – steel; clay – bricks; gypsum – plaster of paris; petroleum – plastics; heavy metals – batteries; gold – jewellery; aluminium – window frames, dinghies; copper – water pipes; gums or plant resins – glues; trees – paper; limestone and clay – cement; silicon – integrated circuits; uranium – yellow cake
- 2 Decaying plant and animal remains
- 3 Over time, as heat and pressure are applied
- 4 Harder and more dense than brown coal
- 5 Anthracite – high lustre, black, looks almost metallic
- 6 At a glance, you can see changing stages in the formation of coal, as well as what contributes to these changes.
- 7 Sun, energy, fossils, oil, millions, petrol, coal, generators, process
- 8 **a** The Sun
b Because they are made from fossils
c Stage 1: Energy from the Sun reaches the Earth.
Stage 2: Plants convert the Sun's energy to food, animals eat the plants, and both plants and animals eventually die.
Stage 3: decaying plant and animal matter is subjected to heat and pressure over time, forming coal.
d Oil: e.g. making plastics, nylon and polyester; lubrication for moving parts of machinery. Petrol: e.g. electricity generation, steel production, cement manufacture.
e Produces carbon dioxide, which contributes to the increasing temperature of the planet.
- 10 Provide figures for the same period of time, in the same format, e.g. all dollar values, not percentages.

Organs and organ systems

- 1 Digestive: stomach, liver, pancreas, small intestines, large intestines
Nervous: brain, spinal cord
Reproductive: ovaries, testes, uterus
Excretory: kidneys, urinary bladder
Respiratory: lungs, bronchi
Circulatory: heart
Skeletal: pelvis, skull, vertebrae

- 2** **a** heat exhaustion, sit in shade, drink water, wet cloths, not able to lose enough heat
b kidney, skin, brain
c not enough feedback to brain
d wear a hat, don't work in the middle of hot days, work in the shade, drink water, rest
- 4** • Remove waste urea from the blood and maintain the correct water balance – Excretory system
 • Change food into water-soluble form so the blood can carry it around the body – Digestive system
 • Add oxygen to the blood so it can be carried to the cells – Respiratory system
 • Manage the interaction of all the systems and all the movement of the body – Nervous system

Particle model of matter

- 2** **a** More
b To increase the rate at which the sugar will dissolve
- 3** Solid: particles just vibrating, particles stay in place, particles jostle each other
 Liquid: particles moving moderately, particles stay in the bottom of the container, particles often collide
 Gas: particles moving rapidly, particles fill the container, particles collide constantly
- 4** Student diagrams should indicate that the state of water is related to its temperature.
- 6** Density increases as the number of particles within a specific volume increases

Solar system

- 1** Mercury: rock; nil; no; 0.4.
 Venus: rock; carbon dioxide; no; 0.7.
 Earth: rock; nitrogen, oxygen, carbon dioxide, other gases; yes; 1.
 Mars: rock; carbon dioxide; yes; 1.5
 Jupiter: gas; hydrogen, helium, methane; yes; 5.
 Saturn: gas; hydrogen, helium, methane; yes; 9.5
 Uranus: gas; hydrogen, helium, methane; yes; 19
 Neptune: gas; hydrogen, helium, methane; yes; 30
- 3** Mercury: 60 million. Venus: 105 million. Earth: 150 million. Mars: 225 million. Jupiter: 750 million. Saturn: 1425 million. Uranus: 2850 million. Neptune: 4500 million
- 4** spins, 24, imaginary, south, daylight, night, east, sunsets, hours, bright, phases, reflected

Technology

6 a

	Force (kg)	distance (cm)	force (kg)	Distance (cm)
A	80	10	10	80
B	60	10	10	60
C	20	20	10	40
D	10	20	10	20

- b** Boltcutters, thick metal
c Scissors, paper

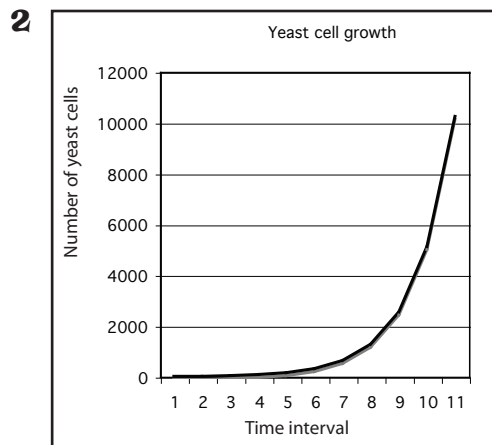
Variables

- 1** Shape of pasta and brand of shampoo are the only items that vary.

Waste from resource use

- 2** (In table order) agricultural lands, untreated sewage, urban roadways, mining leachate, smokestacks and exhaust pipes

Yeast and yoghurt



- 6** Beneficial: yogurt, cheese, bacteria in gut, soil enrichment with legumes
 Harmful: anthrax, whooping cough, pneumonia, bacteria in dental plaque, diphtheria

Zoology

- 1** Fish – shark, perch, barramundi, whiting
 Amphibian – marsh frog, axolotl, turtle, corroboree frog
 Reptile – bluetongue lizard, crocodile
 Bird – parrot, plover, kookaburra
 Mammal – zebra, platypus, quokka
- 2** Fish: ectothermic, scales, moist eggs, gills
 Amphibian: ectothermic, moist skin, lay moist eggs, gills when young, lungs as adult
 Reptile: ectothermic, tough scales, leathery eggs, lungs
 Bird: endothermic, feathers, shelled eggs, lungs
 Mammal: endothermic, hairy skin, live young born, lungs
- 3** Monotremes: pouch, lays eggs, endothermic (warm blood), fur, suckles young
 Marsupials: pouch, placenta, endothermic (warm blood), fur, suckles young
 Placentals: pouch, placenta, endothermic (warm blood), fur, suckles young
- 4** Tiger: large cat, carnivore; India.
 Llama: large camel like herbivore; South America.
 Giraffe: long-necked herbivore; Africa.
 Bilby: nocturnal, small, burrowing marsupial; Australia.
 Hedgehog: small, spiny, placental mammal; UK.
 Bison: large-horned herbivore; North America.

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